



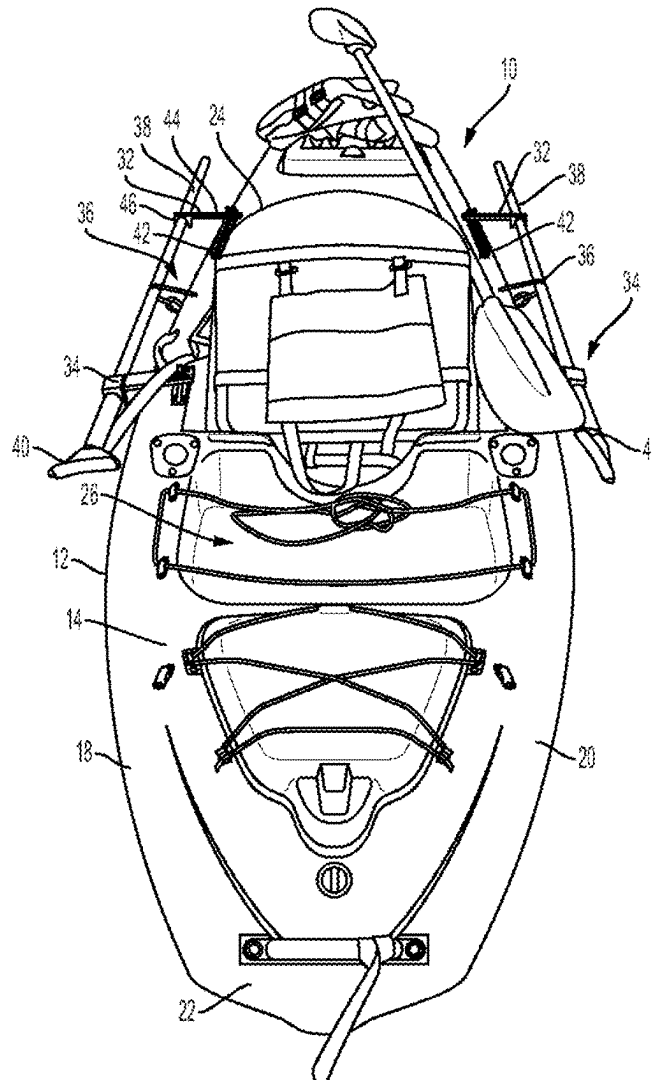
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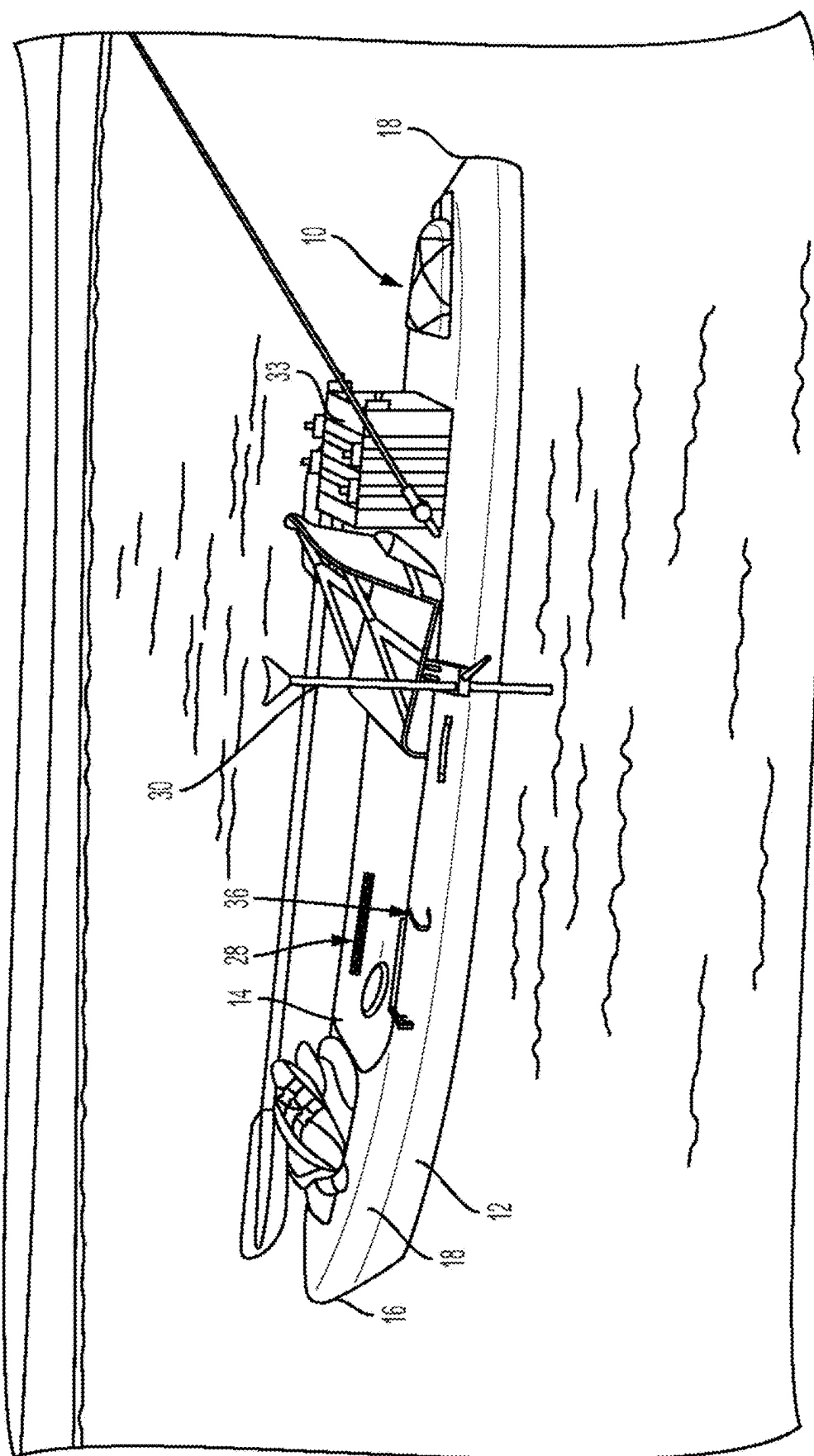
(19) **United States**(12) **Patent Application Publication**
Bennett(10) **Pub. No.: US 2025/0282451 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **ACCESSORY RETENTION SYSTEM FOR WATERCRAFT**(52) **U.S. Cl.**CPC **B63B 32/77** (2020.02); **B63B 32/70** (2020.02); **B63B 34/21** (2020.02)(71) Applicant: **Donny Bennett**, Griffin, GA (US)(72) Inventor: **Donny Bennett**, Griffin, GA (US)(21) Appl. No.: **19/071,372**(22) Filed: **Mar. 5, 2025****Related U.S. Application Data**

(60) Provisional application No. 63/563,097, filed on Mar. 8, 2024, provisional application No. 63/666,056, filed on Jun. 28, 2024.

Publication Classification(51) **Int. Cl.****B63B 32/77** (2020.01)**B63B 32/70** (2020.01)**B63B 34/21** (2020.01)(57) **ABSTRACT**

A retention system for an accessory for use with a watercraft such as a kayak has a bracket configured to be attached to a body of the watercraft and a retainer member attached to the bracket movably between a first orientation with respect to the bracket and a second orientation with respect to the bracket. The retainer member is configured to receive a mounting base, to which a sonar transducer can be mounted. The retainer member is attached to the bracket so that, when the bracket is attached to the body and the mounting base is received by the retainer member, then in the retainer member's first orientation, the mounting base is received by the retainer member so that the sonar transducer is adjacent the elongated body proximate the deck. In the retainer member's second orientation, the mounting base is received so that the sonar transducer is below the hull.





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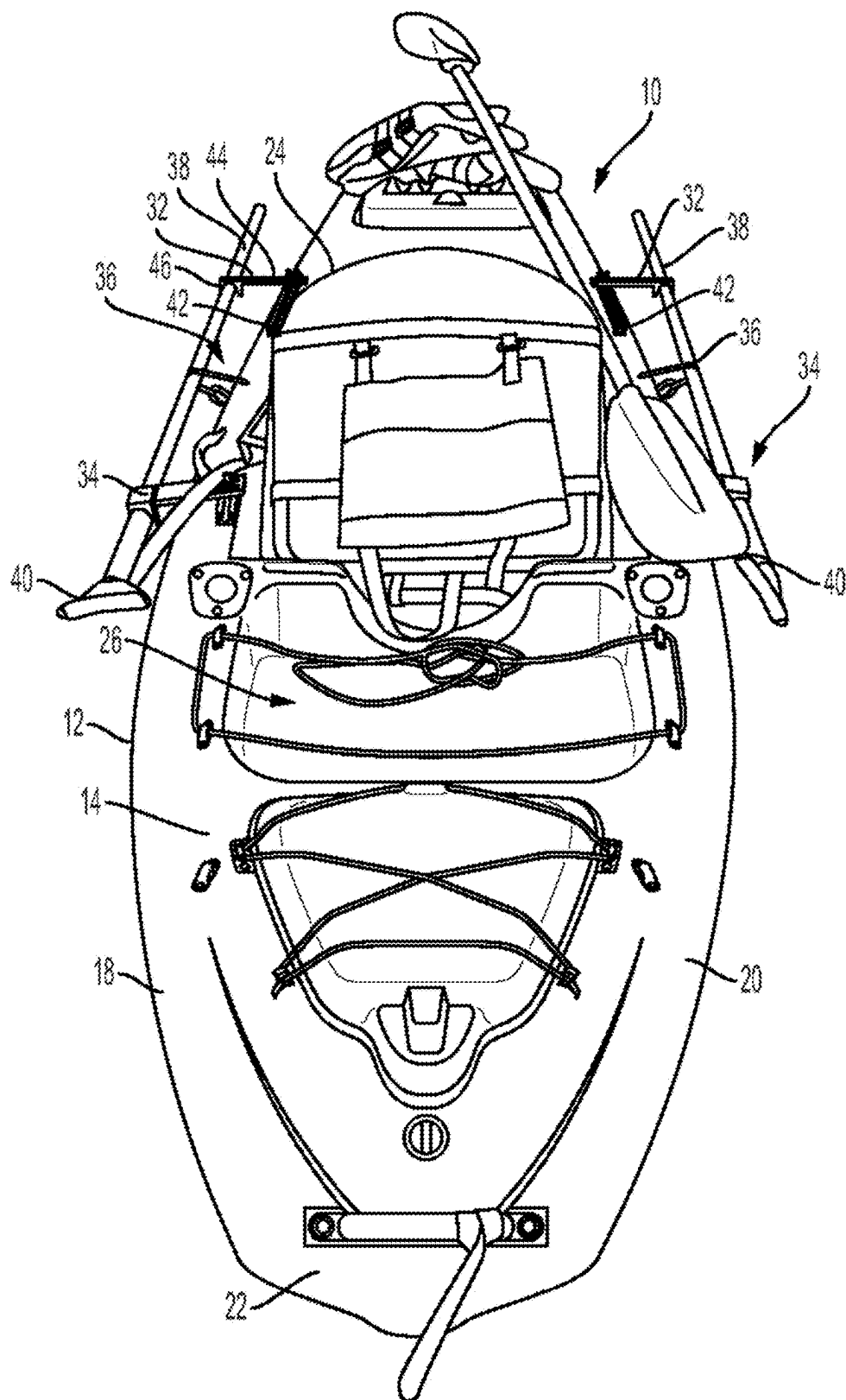


FIG. 2

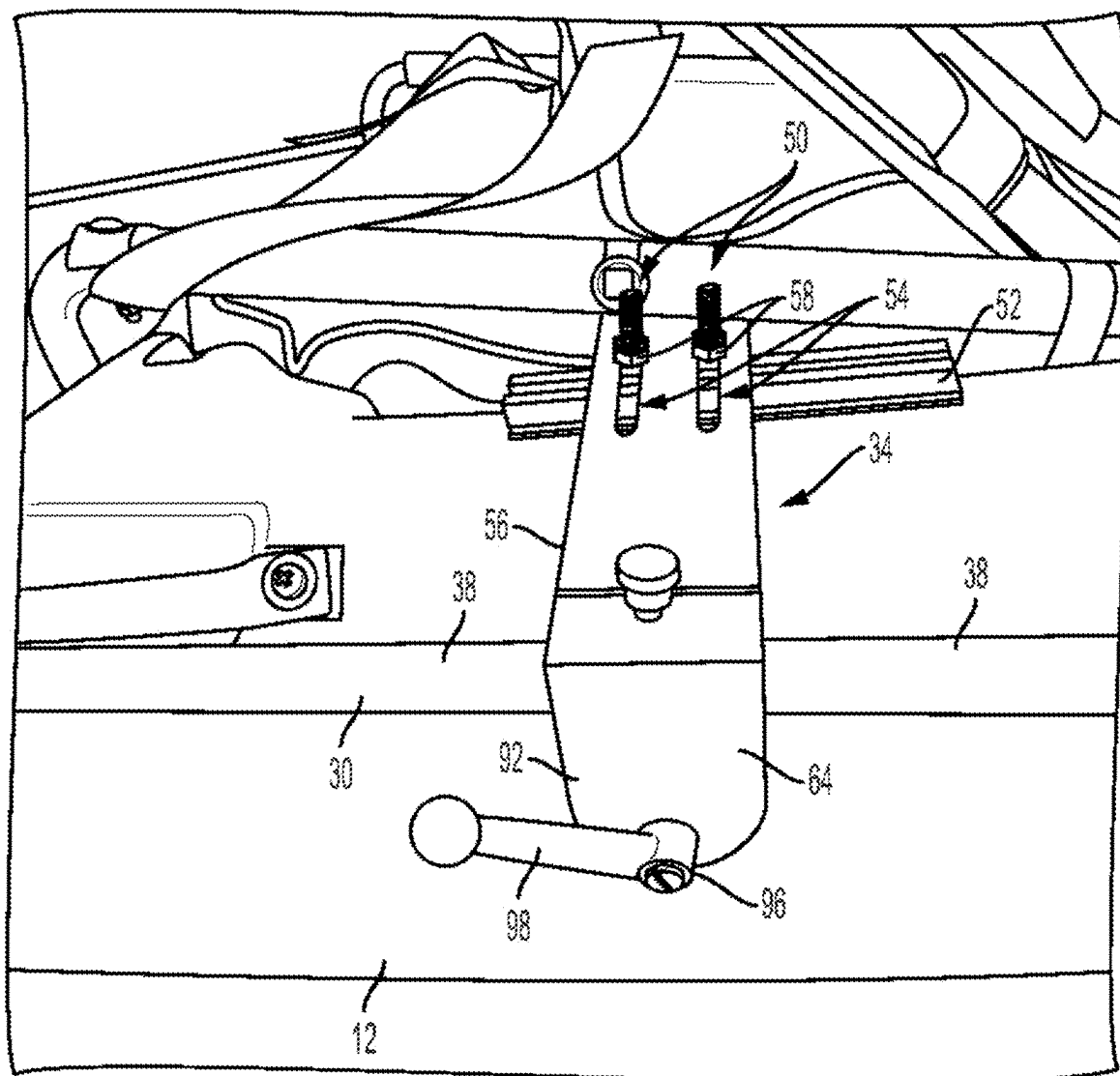


FIG. 3

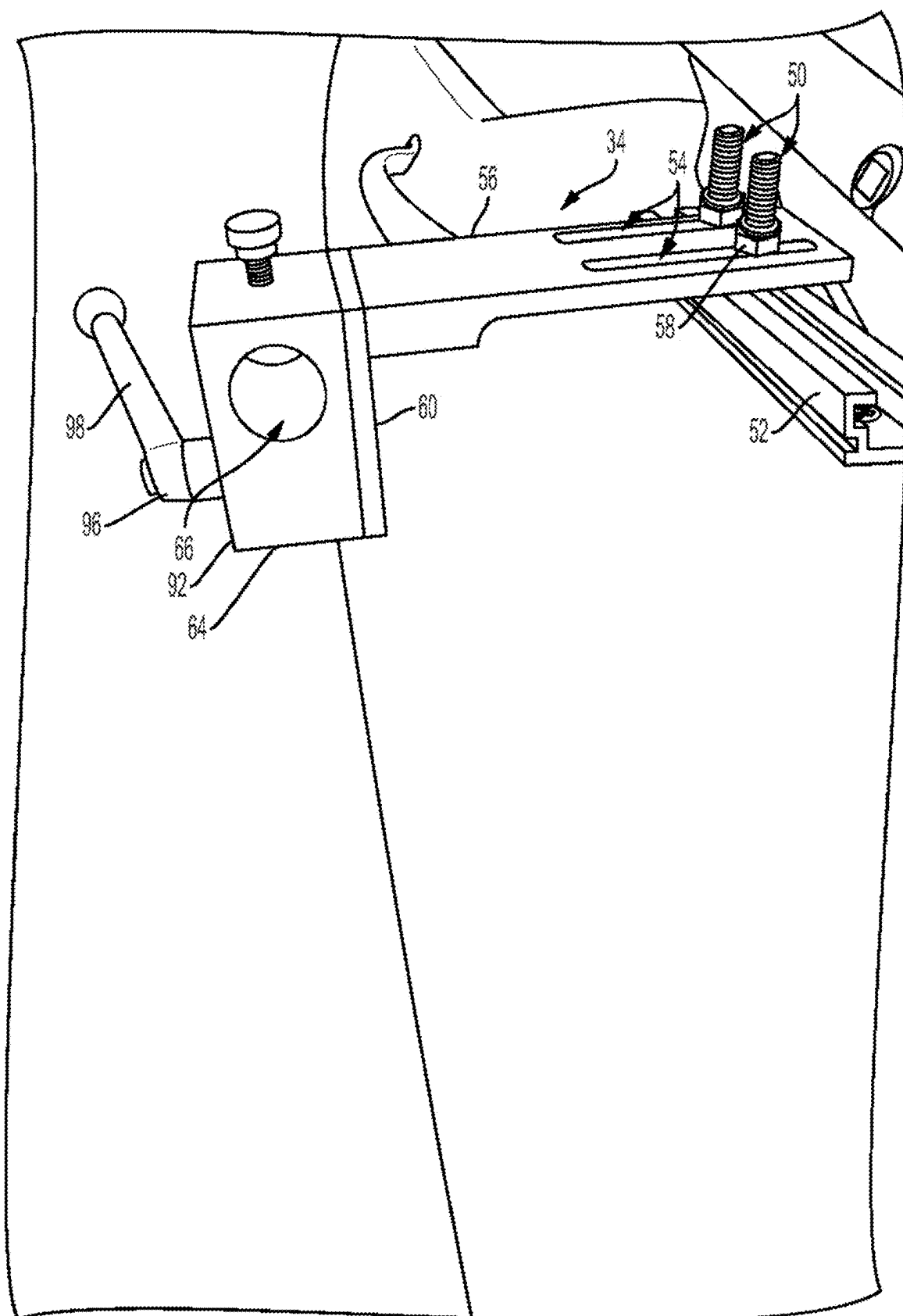


FIG. 4

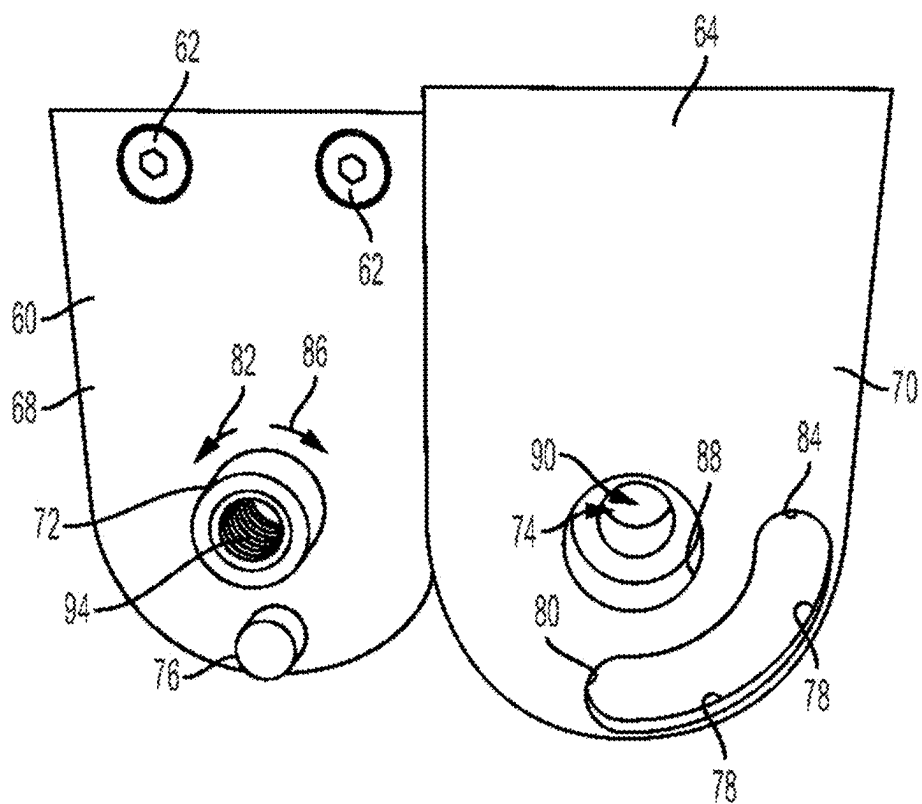


FIG. 5

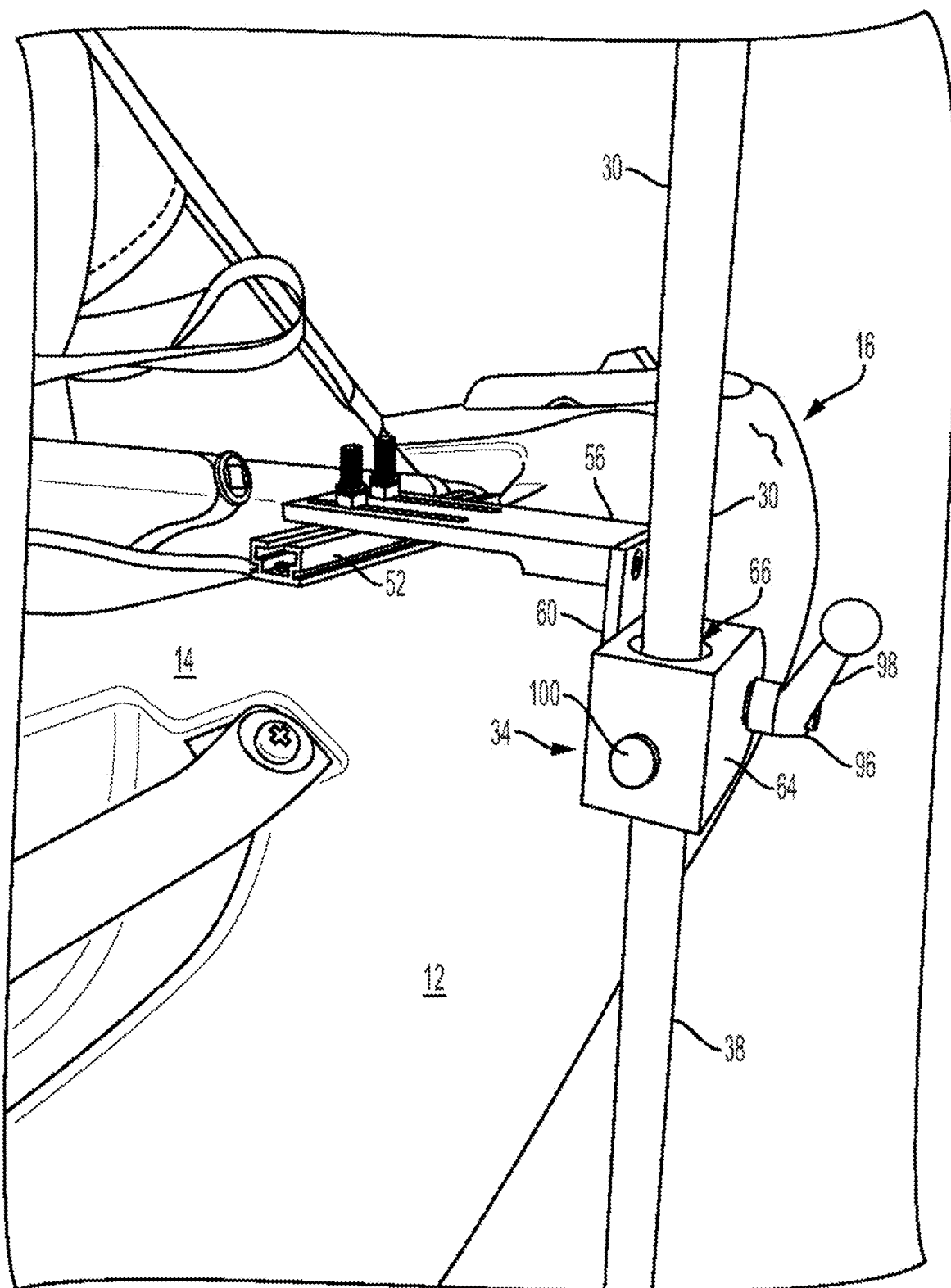


FIG. 6

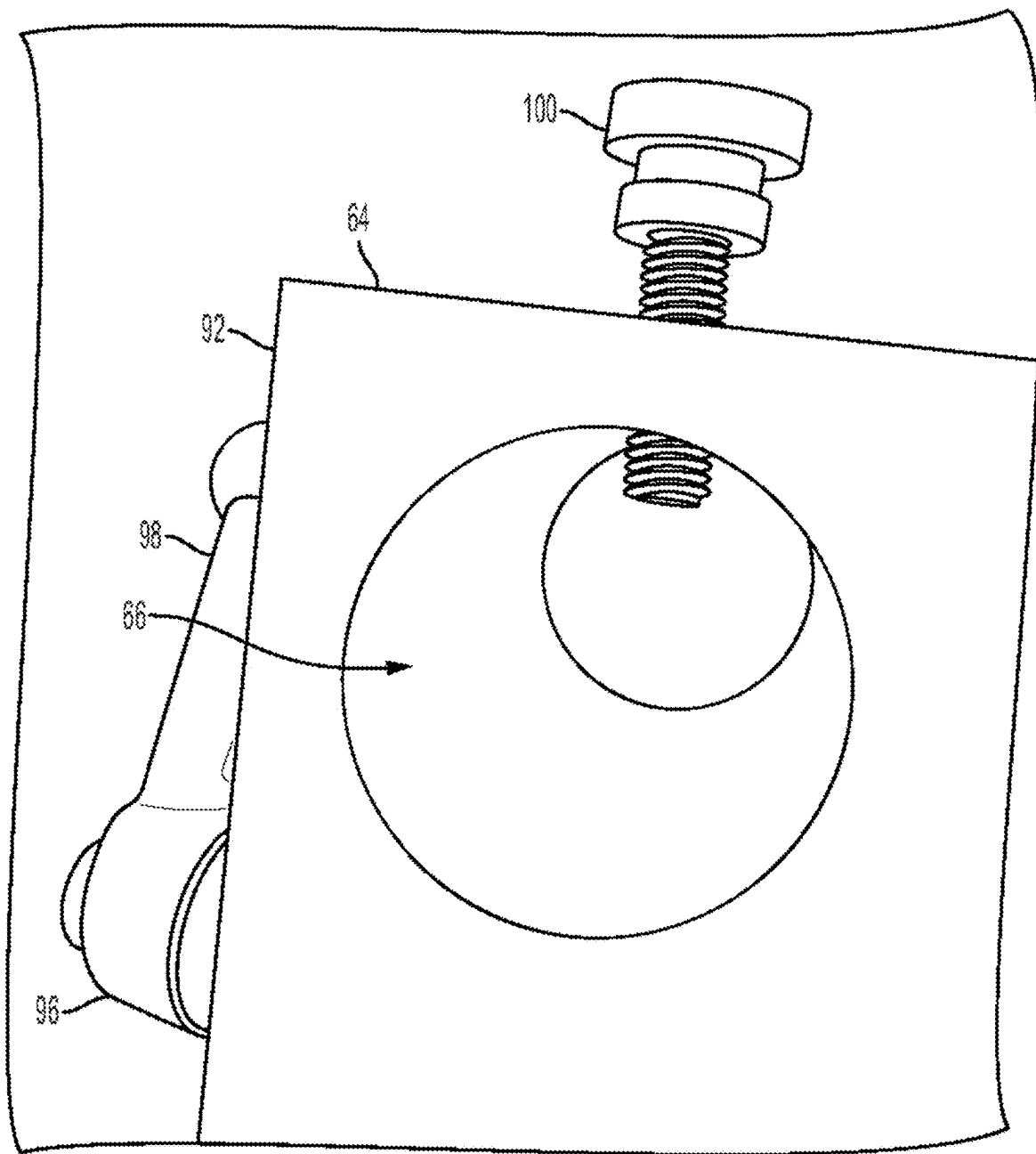


FIG. 7

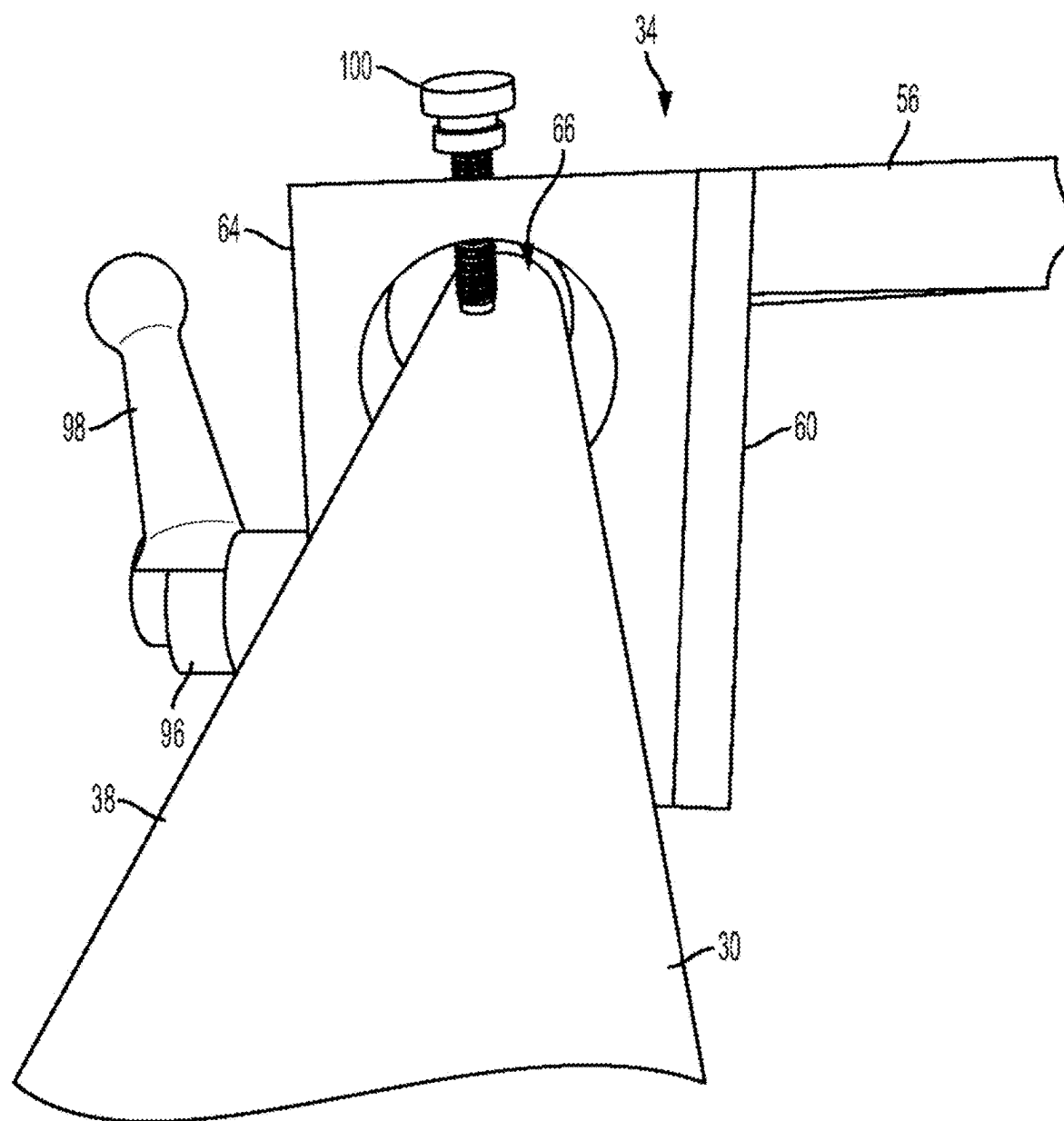


FIG. 8

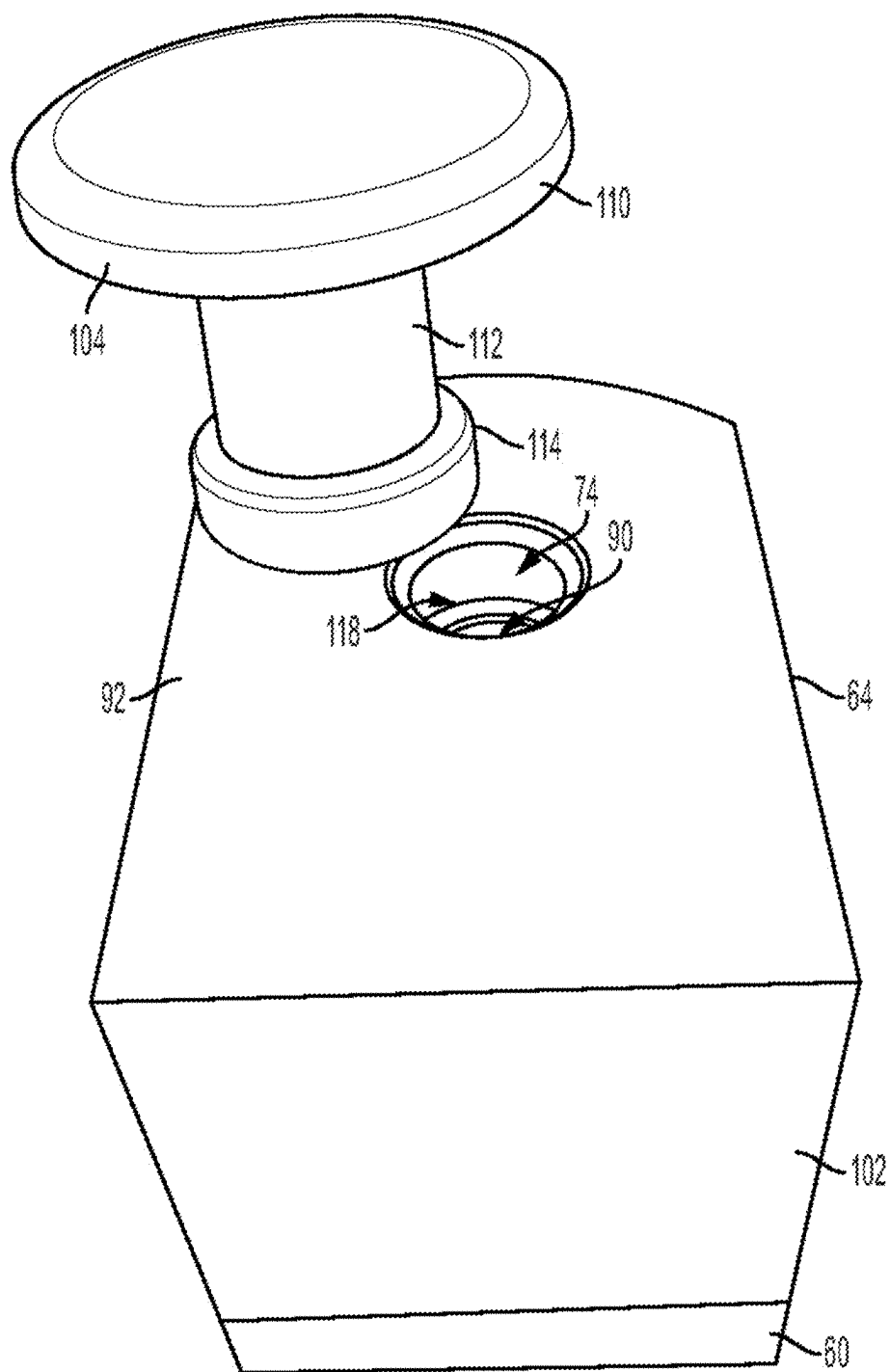


FIG. 9

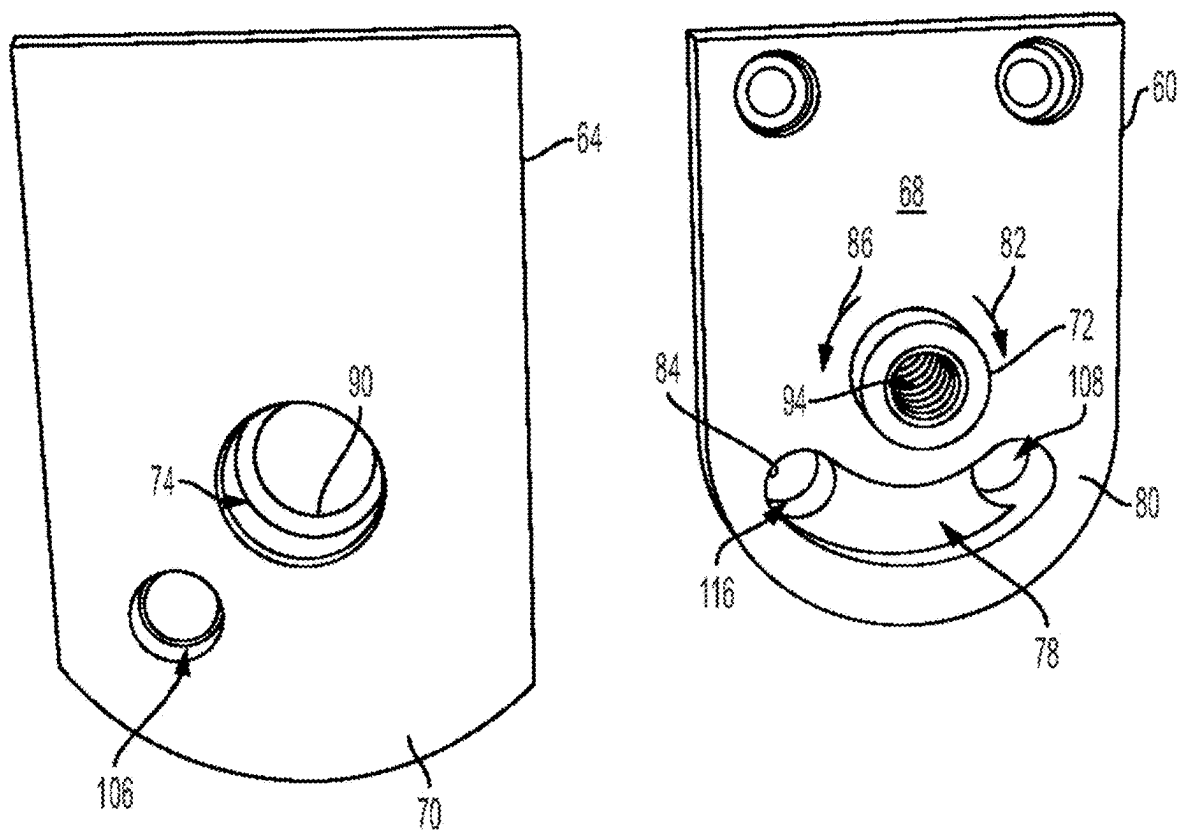


FIG. 10

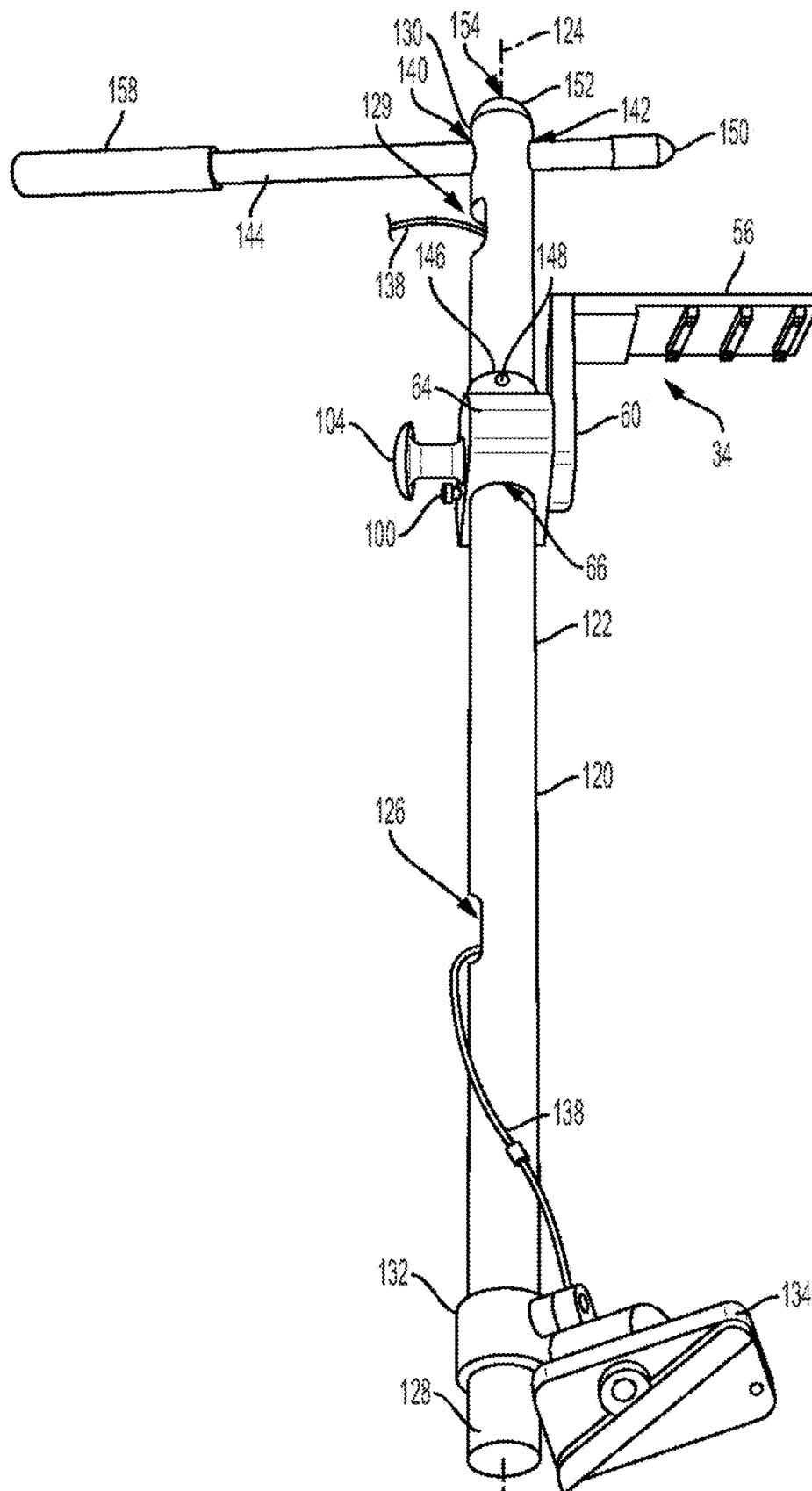


FIG. 11

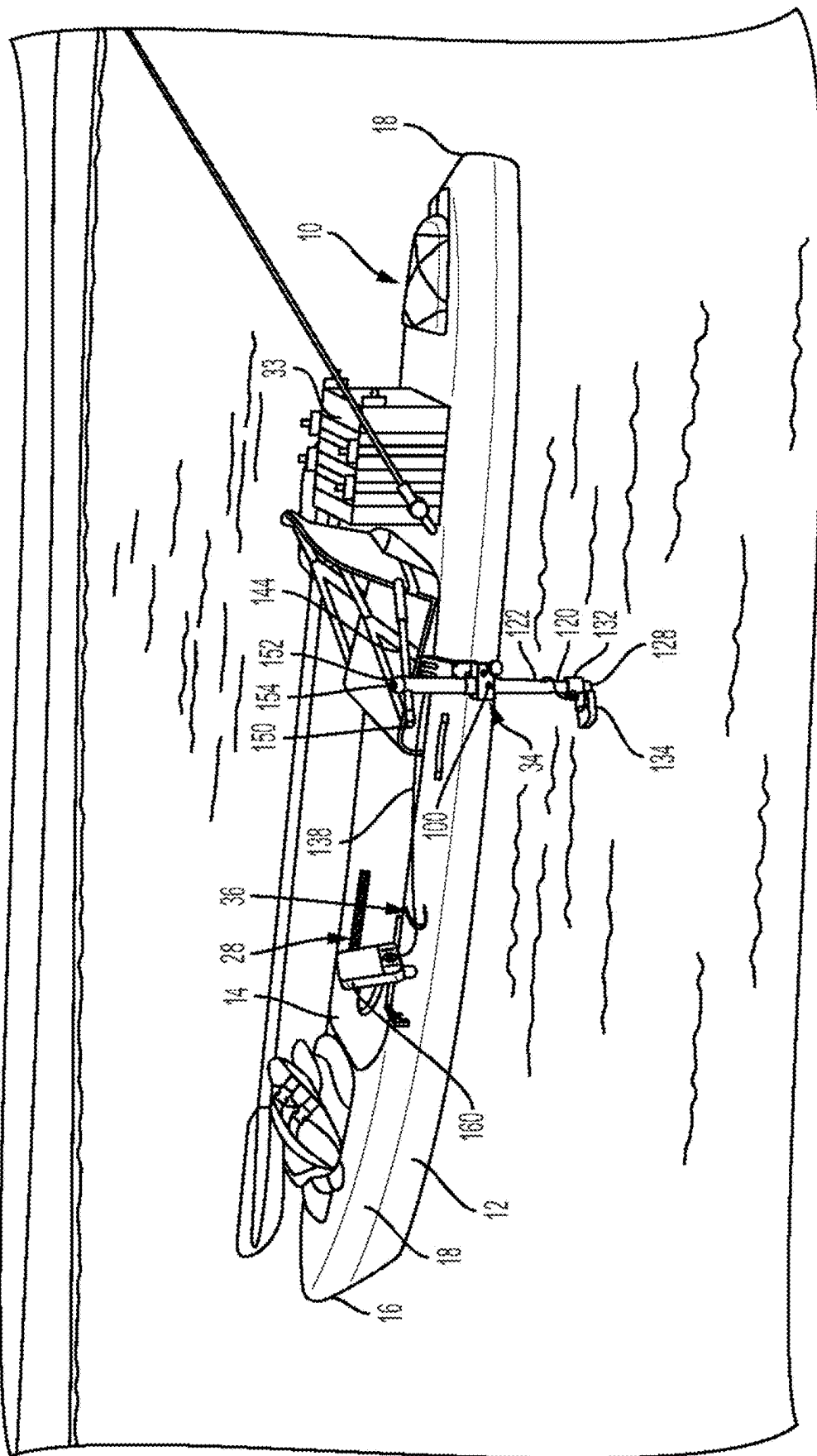


FIG. 12

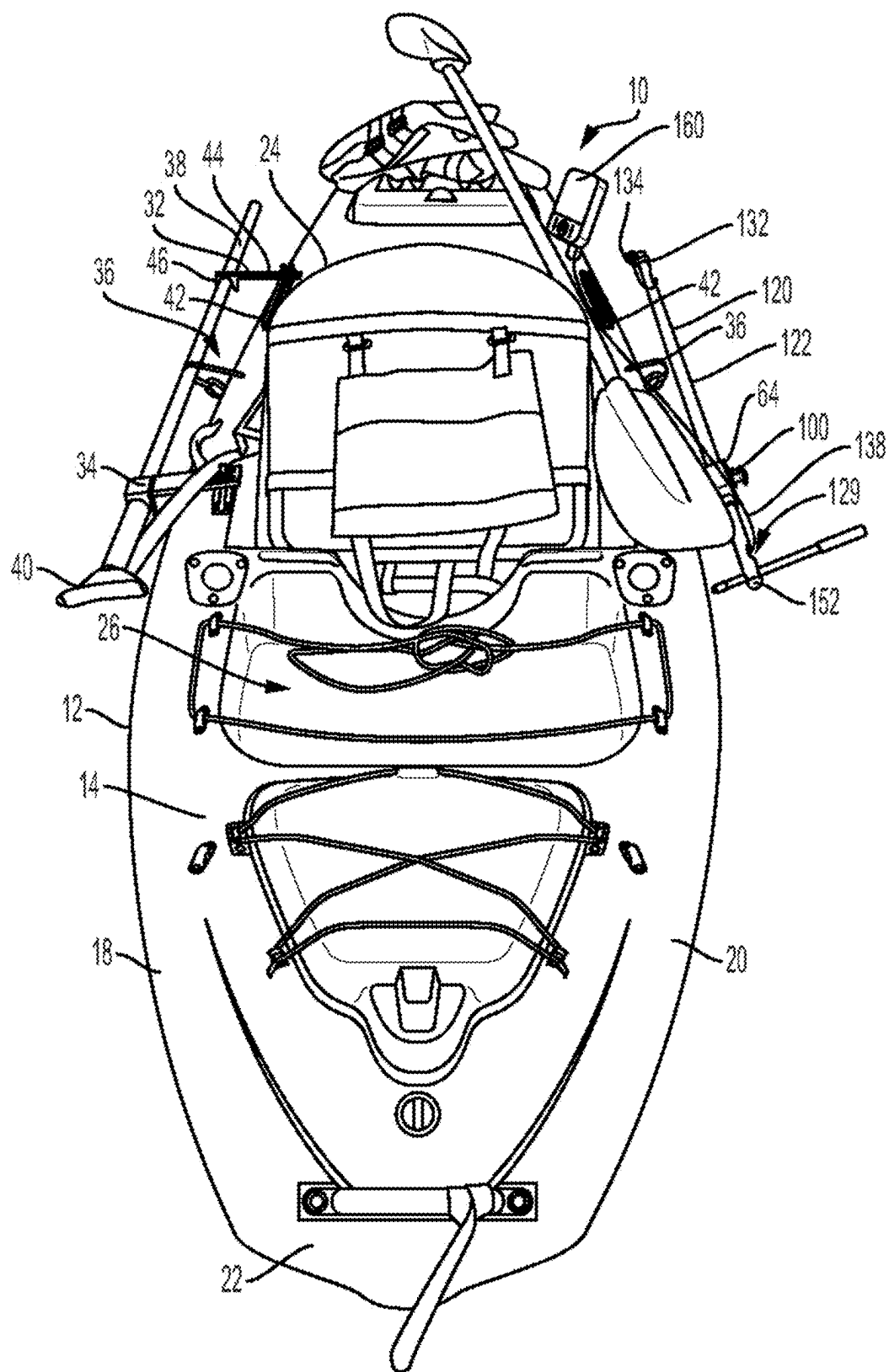


FIG. 13

ACCESSORY RETENTION SYSTEM FOR WATERCRAFT

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims the benefit of U.S. Provisional App. Ser. No. 63/563,097, filed Mar. 8, 2024, entitled “Stake-Out Pole Retention System For Watercraft,” and of U.S. Provisional App. Ser. No. 63/666,056, filed Jun. 28, 2024, entitled “Accessory Retention System For Watercraft.” Both of the foregoing applications are incorporated by reference herein in their entireties for all purposes.

FIELD

[0002] Embodiments relate generally to accessory retention systems for use with watercraft such as fishing kayaks.

BACKGROUND

[0003] Kayaks are small, relatively narrow, elongated watercraft typically powered by paddling of a human operator or, sometimes in the case of fishing-style kayaks, by a small electric trolling motor or by pedal-operated propellers or paddles driven by the operator's feet. Whitewater-type recreational kayaks typically have a single-structure hull that forms the bottom of the craft, with a deck attached to the hull, thereby forming the top of the craft. The single operator sits in the hull, so that the operator's lower body is within the volume defined between the hull and the deck, with the operator's torso extending up through a hole in the deck's center. Typically, the kayak's forward (bow) and rearward (stern) ends come to points so that the kayaks are generally symmetrical about both an axis aligned with the kayak's dimension of elongation and a transverse axis through the center of the craft. In other kayaks, however, including so-called fishing kayaks, the stern may or may not come to a point but, instead, can be blunt as compared to the bow, and the operator sits atop the deck, often on a chair fixed to the deck, which may have a lowered area in front of the chair upon which the operator may stand or kneel. It should be understood, however, that not all fishing kayaks have blunt sterns.

[0004] Operators of fishing kayaks often wish to secure the kayak's position in the water while the operator fishes. Typically, the operator carries with the kayak a long pole (a “stake-out” or “anchor” pole), sometimes about six feet to about ten feet in length, to push down into the bed of the lake or waterway. If the kayak is to be used in deeper waters, longer poles could be used. The kayak may be tethered to the pole, so that the kayak stays within an area around the pole defined by the tether's length. The tether may be secured to the stake-out pole by a ring that loosely surrounds the pole's outer surface and to which the tether is secured so that the ring can move up and down the stake-out pole as the kayak moves up and down (with respect to the stake-out pole, which is pushed into the bed) in response to waves or tide changes.

[0005] Fishing kayaks may include short track rails mounted on one or both sides of the kayak. Each track has a base screwed to the kayak deck, at the deck's edge. A pair of elongated rails extends upward from the base, defining a volume therebetween with a dimension of elongation generally parallel to the elongated kayak body's dimension of elongation. At the top of the rails, short, horizontal flanges

extend inward toward each other but stop short of each other to thereby define a slit between them that is narrower than space below (extending between the elongated vertical rails). The heads of one or more T-bolts are received in the volume between the vertical rails, so that the bolts' threaded distal ends extend upward through the slit. Respective threaded nuts threadedly receive the bolts' threaded distal ends and can be tightened down to bring the heads of the T-bolts up into engagement with the flanges' bottom surfaces, so that as the nut is screwed down into engagement with the flanges' top surfaces, the nut and the T-bolt head engage the flanges to thereby hold the T-bolt in place in the track. By loosening the nut, the operator can move the T-bolt in the track in either direction in the slit. The T-bolts can be used to secure accessories, such as cameras or depth finders, to the kayak and are sometimes referred to as universal track rails.

[0006] In one mechanism to secure a stake-out pole, a triangular bracket having respective holes at two of the triangle's corners receives the threaded ends of two T-bolts whose T-shaped heads are received in the track rail. Hand actuatable nuts are received over the threaded ends, enabling the operator to secure the bracket onto the kayak edge. The third end of the bracket extends outward of the kayak body to one side, and a vertically oriented through hole slidably receives the stake-out pole. The operator inserts the stake-out pole vertically through the outboard hole in the bracket and pushes the pole into the water body bed, and the loose fit between the bracket and the pole allows the kayak to move vertically with waves or tide change while retaining the kayak laterally with respect to the pole. The bracket may have an arcuate slot extending vertically through the bracket and in communication with one of the two inboard holes at the other bracket corners. When the pole is not received by the bracket, the two hand actuatable nuts can be loosened and the bracket pivoted about one of the T-bolts so that the outboard portion of the bracket can be pivoted inboard over the deck edge as the arcuate groove accepts, and allows that end of the triangular bracket to move with respect to, the other T-bolt.

[0007] Fishing kayaks may also have anchor trolleys, comprised of a metal ring attached to opposing ends of a single length of cable or rope. The rope loops around two pulleys, one at the kayak bow and the other at the stern, so that the rope extends in a long loop adjacent the hull, just beneath one side edge of the deck. The ring is part of that loop and is on its outward half. To anchor the kayak, the operator inserts the stake-out pole vertically through the ring and pushes the pole into the water body bed. The loose fit between the ring and the pole allows the kayak to move vertically with waves or tide change while retaining the kayak laterally with respect to the pole. The operator can further control the kayak's position with respect to the pole by reaching down over the side of the kayak, manually gripping the interior half of the rope loop, and pulling the inward rope away from either the bow or the stern, thereby causing the outer rope half to move, with respect to the kayak, in the other direction via the rope loop's movement about the pulleys. Since the rope's position is stationary at the ring with respect to the stake-out pole, this moves the kayak with respect to the pole. To achieve greater mobility relative to the pole, it is known to connect the pole to the ring on the anchor trolley by a tether.

[0008] It is known to use a hydraulic or electric motor to drive a stake-out pole into the water bed. More generally, it is also known to simply connect the anchor pole to the kayak with a tether, thereby allowing the kayak to move with respect to the pole about and within a circle centered at the pole with a radius of the tether's length.

[0009] It is known to use sonar devices on kayaks to locate fish in the volume of water beneath the watercraft. Kayak manufacturers often form the molded kayak hull with a recessed pocket in which is disposed a mounting device to which the sonar transducer may attach. A sealed hole allows electronic wiring to pass from the transducer up into the kayak hull interior and, then up to the kayak deck, where it connects to a display unit. It is also known to mount the sonar transducer entirely inside the hull, to the interior side of the hull bottom so that the transducer directs its signal through the hull. It is known to attach the transducer to a fixed bracket, which is carried in the kayak until the fish finder is needed. At that point, the operator secures the fixed bracket on the side of the kayak hull, so that the bracket extends the transducer down into the water.

[0010] The present invention may recognize and address one or more considerations of prior art constructions and methods, as recited above or otherwise.

BRIEF SUMMARY

[0011] A watercraft according to one or more embodiments of the present disclosure has an elongated body with a forward end, a rearward end opposite the forward end along the body's dimension of elongation, a hull extending from the forward end to the rearward end and forming a bottom of the body, and a deck extending from the forward end to the rearward and forming a top of the body. A bracket is attached to the body. A retainer member is attached to the bracket movably between a first orientation with respect to the bracket and a second orientation with respect to the bracket. An elongated stake-out pole received by the retainer member. The bracket is attached to the body and the retainer member is attached to the bracket so that, in the retainer member's first orientation, the stake-out pole's dimension of elongation is generally parallel to the body's dimension of elongation, and in the retainer member's second orientation, the stake-out pole's dimension of elongation is generally perpendicular to the body's dimension of elongation, and in the retainer member's second orientation, the stake-out pole is outward of a side of the body extending between the forward end and the rearward end.

[0012] In one or more other embodiments, a kayak has an elongated body with a forward end, a rearward end opposite the forward end along the body's dimension of elongation, a hull extending from the forward end to the rearward end and forming a bottom of the body, and a deck extending from the forward end to the rearward and forming a top of the body. A chair is attached to the deck. A bracket is attached to the deck and extends across and outward of a perimeter of the body defined by the forward end, the rearward end, and opposing sides of the body at one of the sides of the body, wherein each opposing side extends between the forward end and the rearward end. A collar is disposed at least partially outward of the perimeter and attached to the bracket movably between a first orientation with respect to the bracket and a second orientation with respect to the bracket. An elongated stake-out pole is received by the collar so that the collar at least partially

encloses an outer perimeter of the stake-out pole. The bracket is attached to the body, and the collar is attached to the bracket, so that, in the collar's first orientation, the stake-out pole's dimension of elongation is generally parallel to the body's dimension of elongation, in the collar's second orientation, the stake-out pole's dimension of elongation is generally perpendicular to the body's dimension of elongation, and in the collar's second orientation, the stake-out pole is outward of the one of the sides of the body extending between the forward end and the rearward end. A selectively engageable first lock is between the collar and the stake-out pole. A selectively engageable second lock is between the bracket and the collar. The second lock is configured to resist movement of the collar out of the first orientation and to resist movement of the collar out of the second orientation.

[0013] In one or more still further embodiments, a retention system for a stake-out pole for use with a kayak having an elongated body with a forward end, a rearward end opposite the forward end along the body's dimension of elongation, a hull extending from the forward end to the rearward end and forming a bottom of the body, and a deck extending from the forward end to the rearward end and forming a top of the body, has a bracket configured to be attached to the body and a retainer member attached to the bracket movably between a first orientation with respect to the bracket and a second orientation with respect to the bracket, wherein the retainer member is configured to receive an elongated stake-out pole. The retainer member is attached to the bracket so that, when the bracket is attached to the body and the elongated stake-out pole is received by the retainer member, then in the retainer member's first orientation, the stake-out pole's dimension of elongation is generally parallel to the body's dimension of elongation, and in the retainer member's second orientation, the stake-out pole's dimension of elongation is generally perpendicular to the body's dimension of elongation, and in the retainer member's second orientation, the stake-out pole is outward of a side of the body extending between the forward end and the rearward end.

[0014] In another embodiment, a watercraft has an elongated body with a forward end, a rearward end opposite the forward end along the body's dimension of elongation, a hull extending from the forward end to the rearward end and forming a bottom of the body, and a deck extending from the forward end to the rearward and forming a top of the body. A bracket is attached to the body. A retainer member is attached to the bracket movably between a first orientation with respect to the bracket and a second orientation with respect to the bracket. A mounting base is received by the retainer member. A sonar transducer is mounted at a distal end of the mounting base. The bracket is attached to the body, and the retainer member is attached to the bracket so that, in the retainer member's first orientation, the mounting base is received by the retainer member so that the sonar transducer is adjacent the elongated body proximate the deck. In the retainer member's second orientation, the mounting base is received by the retainer member so that the sonar transducer is below the deck. In the retainer member's second orientation, the mounting base is outward of a side of the body extending between the forward end and the rearward end.

[0015] In a still further embodiment, a retention system for an accessory for use with a kayak having an elongated body

with a forward end, a rearward end opposite the forward end along the body's dimension of elongation, a hull extending from the forward end to the rearward end and forming a bottom of the body, and a deck extending from the forward end to the rearward end and forming a top of the body, has a bracket configured to be attached to the body and a retainer member attached to the bracket movably between a first orientation with respect to the bracket and a second orientation with respect to the bracket. The retainer member is configured to receive a mounting base. A sonar transducer is mounted at a distal end of the mounting base. The retainer member is attached to the bracket so that, when the bracket is attached to the body and the mounting bracket is received by the retainer member, in the retainer member's first orientation, the mounting base is received by the retainer member so that the sonar transducer is adjacent the elongated body proximate the deck. In the retainer member's second orientation, the mounting base is received by the retainer member so that the sonar transducer is below the deck. In the retainer member's second orientation, the mounting base is outward of a side of the body extending between the forward end and the rearward end.

[0016] In another embodiment, a kayak has an elongated body comprising a forward end, a rearward end opposite the forward end along the body's dimension of elongation, a hull extending from the forward end to the rearward end and forming a bottom of the body, and a deck extending from the forward end to the rearward end and forming a top of the body. A chair is attached to the deck. A bracket is attached to the body. A retainer member is attached to the bracket movably between a first orientation with respect to the bracket and a second orientation with respect to the bracket. A mounting base is received by the retainer member. A sonar transducer is mounted at a distal end of the mounting base. The bracket is attached to the body, and the retainer member is attached to the bracket, so that in the retainer member's first orientation, the mounting base is received by the retainer member so that the sonar transducer is adjacent the elongated body proximate the deck, so that in the retainer member's second orientation, the mounting base is received by the retainer member so that the sonar transducer is below the deck, and so that in the retainer member's second orientation, the mounting base is outward of a side of the body extending between the forward end and the rearward end.

[0017] In a further embodiment, a kayak has an elongated body comprising a forward end, a rearward end opposite the forward end along the body's dimension of elongation, a hull extending from the forward end to the rearward end and forming a bottom of the body, and a deck extending from the forward end to the rearward end and forming a top of the body. A chair is attached to the deck. A bracket is attached to the body. A retainer member is attached to the bracket movably between a first orientation with respect to the bracket and a second orientation with respect to the bracket. An elongated pole is received by the retainer member. The bracket is attached to the body, and the retainer member is attached to the bracket, so that, in the retainer member's first orientation, the pole is received by the retainer member so that the pole's dimension of elongation is generally parallel to the body's dimension of elongation and, in the retainer member's second orientation, the pole's dimension of elongation is generally perpendicular to the body's dimension of elongation.

[0018] In accordance with another embodiment, the present invention provides a retention system for use with a watercraft. The watercraft comprises an elongated body comprising a forward end, a rearward end opposite the forward end along the body's dimension of elongation, a hull extending from the forward end to the rearward end and forming a bottom of the body, and a deck extending from the forward end to the rearward end and forming a top of the body. The retention system comprises a bracket configured to be attached to the body and a retainer member attached to the bracket movably between a first orientation with respect to the bracket and a second orientation with respect to the bracket. The retention system also comprises a mounting base received by the retainer member, the mounting base having a proximate end and an opposite distal end. When the bracket is attached to the body and the retainer member is attached to the bracket, in the retainer member's first orientation, the mounting base is received by the retainer member so that the mounting base distal end is adjacent the elongated body proximate the deck, in the retainer member's second orientation, the mounting base is received by the retainer member so that the mounting base distal end is below the deck, and in the retainer member's second orientation, the mounting base is outward of a side of the body extending between the forward end and the rearward end.

[0019] In a further embodiment, the present invention provides a retention system for use with a watercraft. The watercraft has an elongated body with a forward end, a rearward end opposite the forward end along the body's dimension of elongation, a hull extending from the forward end to the rearward end and forming a bottom of the body, and a deck extending from the forward end to the rearward end and forming a top of the body. The retention system comprises a bracket configured to be attached to the body and a retainer member attached to the bracket movably between a first orientation with respect to the bracket and a second orientation with respect to the bracket. The retainer member is configured to receive an elongated pole. The retention system also comprises at least one watercraft accessory coupled with a distal end of the elongated pole. The retainer member is attached to the bracket so that, when the bracket is attached to the body and the elongated pole is received by the retainer member, in the retainer member's first orientation, the elongated pole is received by the retainer member so that the elongated pole is adjacent the elongated body proximate the deck, in the retainer member's second orientation, the elongated pole is received by the retainer member so that the elongated pole's dimension of elongation is generally perpendicular to the body's dimension of elongation, and in the retainer member's second orientation, the mounting base is outward of a side of the body extending between the forward end and the rearward end.

[0020] Yet another embodiment provides retention system for use with a watercraft. The watercraft comprises an elongated body comprising a forward end, a rearward end opposite the forward end along the body's dimension of elongation, a hull extending from the forward end to the rearward end and forming a bottom of the body, and a deck extending from the forward end to the rearward end and forming a top of the body. The retention system comprises a bracket configured to be attached to the body and a retainer member attached to the bracket movably between a first orientation with respect to the bracket and a second orientation

tation with respect to the bracket. The retainer member is configured to receive an elongated pole. When the bracket is attached to the body, the retainer member is attached to the bracket, and the elongated pole is received in the retainer member, in the retainer member's first orientation, the pole is received by the retainer member so that the pole's dimension of elongation is generally parallel to the body's dimension of elongation, and in the retainer member's second orientation, the pole's dimension of elongation is generally perpendicular to the body's dimension of elongation.

[0021] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate one or more embodiments of the invention and, together with the description, serve to explain the principles of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended drawings, in which:

[0023] FIG. 1 is a perspective view of a kayak in accordance with an embodiment of the present invention, disposed on a water body;

[0024] FIG. 2 is a rear perspective view of the kayak as in FIG. 1;

[0025] FIG. 3 is a perspective view of an embodiment of a watercraft accessory retainer, retaining a stake-out pole, on the kayak as in FIG. 1;

[0026] FIG. 4 is a perspective view of the accessory retainer as in FIG. 3 on the kayak as in FIG. 1;

[0027] FIG. 5 is a disassembled plan view of a retainer member and bracket end plate of the accessory retainer as in FIGS. 3 and 4;

[0028] FIG. 6 is a perspective view of the accessory retainer as in FIG. 3, retaining a stake-out pole, on the kayak as in FIG. 1;

[0029] FIG. 7 is a partial side view of the accessory retainer as in FIG. 3;

[0030] FIG. 8 is a partial side view of the accessory retainer as in FIG. 3, retaining a stake-out pole;

[0031] FIG. 9 is a perspective view of a retainer member and bracket end plate of an embodiment of an accessory retainer for use with a kayak as in FIG. 1;

[0032] FIG. 10 is a disassembled plan view of the retainer member and bracket end plate of FIG. 9;

[0033] FIG. 11 is a perspective view of an embodiment of a watercraft accessory retainer according to an embodiment of the present invention, retaining a sonar transducer, for use on the kayak as in FIG. 12;

[0034] FIG. 12 is a perspective view of a kayak in accordance with an embodiment of the present invention, disposed on a water body; and

[0035] FIG. 13 is a rear perspective view of the kayak as in FIG. 12.

[0036] Repeat use of reference characters in the present specification and drawings is intended to represent same or analogous features or elements of the invention according to the disclosure.

DETAILED DESCRIPTION

[0037] Reference will now be made in detail to presently preferred embodiments of the invention, one or more examples of which are illustrated in the accompanying drawings. Each example is provided by way of explanation, not limitation, of the invention. In fact, it will be apparent to those skilled in the art that modifications and variations may be made in the present invention without departing from the scope and spirit thereof. For instance, features illustrated or described as part of one embodiment may be used on another embodiment to yield a still further embodiment. Thus, it is intended that the present disclosure covers such modifications and variations as come within the scope of the appended claims and their equivalents. All values indicated below are intended to be approximated values.

[0038] It should be understood that terms of orientation, e.g., "forward," "rearward," "upper," "lower," and similar terms as used herein are intended to refer to relative orientation of components of the devices described herein with respect to each other under an assumption of a consistent point of reference but do not require any specific orientation of the overall system. Thus, for example, the discussion herein may refer to the "forward," "rearward," "lateral," "side," or similar descriptions, referring to areas of or directions with respect to a watercraft. Such terms may be used in the present disclosure and claims and will be understood to refer to a relative orientation but not to an orientation of a claimed device with respect to an external frame of reference.

[0039] Further, the term "or" as used in this application and the appended claims is intended to mean an inclusive "or" rather than an exclusive "or." That is, unless specified otherwise, or clear from the context, the phrase "X employs A or B" is intended to mean any of the natural inclusive permutations. That is, the phrase "X employs A or B" is satisfied by any of the following instances: X employs A; X employs B; or X employs both A and B. In addition, the articles "a" and "an" as used in this application and the appended claims should generally be construed to mean "one or more" unless specified otherwise or clear from the context to be directed to a singular form. Throughout the specification and claims, the following terms take at least the meanings explicitly associated herein, unless the context dictates otherwise. The meanings identified below do not necessarily limit the terms, but merely provide illustrative examples for the terms. The meaning of "a," "an," and "the" may include plural references, and the meaning of "in" may include "in" and "on." The phrase "in one embodiment," as used herein does not necessarily refer to the same embodiment, although it may. The phrase "at least one of A and B" is satisfied by any of A alone, B alone, A and B alone, and A and B with others. The phrase "one of A and B" is satisfied by A, whether or not also in the presence of B, and by B, whether or not also in the presence of A.

[0040] Various aspects or features will be presented in terms of systems that may include a number of devices, components, modules, and the like. It is to be understood and appreciated that the various systems may include additional devices, components, modules, etc. and/or may not include all of the devices, components, modules etc. discussed in connection with the figures. A combination of these approaches may also be used.

[0041] Referring to FIGS. 1 and 2, a kayak 10 has a body formed by a hull 12 and a deck 14, each of which may be

formed from various materials, such as fiberglass, thermoplastic polymers such as acrylonitrile butadiene styrene (ABS), or aromatic polyamide fiber. While the hull and deck may be formed as a continuous-material structure, as illustrated in the Figures, it should also be understood that the kayak can be formed of discrete panels (e.g., a hull panel and a deck panel) attached by fasteners or adhesive or other constructions. Hull 12 forms the bottom of the body and defines the interface between the body and the body of water when the kayak is in use but may also be considered to include the structure that extends upward from the waterline to the body edges that turn generally horizontally inward to form the kayak's top surface, or deck, 14.

[0042] The body of kayak 10 has a bow 16 that is defined as a ridge formed by the intersection of two opposing sides 18 and 20 of the hull and deck. Those opposing sides extend rearward from the bow to a blunted stern 22. A chair 24 is attached to deck 14 at a middle portion thereof. Chair 24 may be attached to the deck in a fixed orientation, with the sitting position thereof facing toward bow 16 or with capability to swivel side to side from the forward-facing orientation. The deck defines vertically recessed portions 26 and 28, respectively aft and fore of chair 24. Recessed portion 26 may receive accessory gear, such as an ice chest 33, while recessed portion 28 provides a surface upon which to rest the user's feet when the user sits in chair 24 or on which to stand or kneel. Thus, as should be understood in this art, the deck may not be a level surface and may have portions thereof that are lower than the deck's highest surface.

[0043] As illustrated in FIG. 2, kayak 10 includes a pair of stake-out poles 30, each disposed on a respective side 18/20 of the kayak body by respective pairs of retainers 32 and 34 and respective tethers 36. Each stake-out pole has an elongated rigid shaft 38 and a handle 40 at one end of the shaft, with the pole being approximately six feet long in its dimension of elongation, although other lengths may be used. The shaft of each pole may be made of any suitable material, such as fiberglass, aluminum, or polyvinyl chloride (PVC).

[0044] At the kayak body's forward end, each aluminum front retainer 32 is secured to the kayak by one or two T-bolts extending upward from a universal track rail 42 secured to forward edge 18 or 20 of body deck 14. The (or each) T-bolt shaft extends upward through a respective slot (in one or more embodiments, in which there are two slots, the slots are elongated and parallel to each other, similar to the slots 54 in bracket 56, FIG. 3, discussed below) in an elongated bracket portion 44 of retainer 32. A threaded nut is threaded down onto the upward-facing threaded distal end of the T-bolt shaft and is tightened onto the bolt thread to engage the top of bracket portion 44 and thereby secure retainer 32 to the kayak body. A downward-facing yoke portion 46 of retainer 32 is formed by two downward-depending flanges that define therebetween a downward-opening slot that receives stake-out pole shaft 38.

[0045] As apparent in FIG. 2, the slot of yoke portion 46 of retainer 32 is sufficiently wide to provide a sliding fit for pole shaft 38 but to also engage the shaft. Tether 36 has a looped upper end that is attached to the kayak deck edge by a cleat or to small handle strap fixed to the deck by one or more screws or pop rivets and, at its lower end, to a carabiner or other manually operable clip, which is itself attached to the kayak body, e.g. by screws or pop rivets, or by attachment to such a handle strip that is itself attached to the kayak

body by such means. As should be understood, such tethers are sometimes provided on the kayak by the manufacturer to store related gear. To secure the tether about the pole shaft, the operator grips the tether's loose lower end, wraps the tether around the pole shaft, and secures a loop at the tether's lower end into the carabiner or, if the carabiner is fixed to the tether and selectively attached to the watercraft body by a handle strap that is fixed to the body, secures the carabiner to the handle strap. The tether's length is short enough to bias the pole shaft inward toward the kayak body, more tightly against the inward flange of yoke portion 46 and the inner surface of bore 66 (discussed below) in retainer member 34, to thereby inhibit the pole's vibrations in its horizontal position as shown in FIG. 2 as the operator moves the kayak over a water surface. The tethers' engagement of the stake-out poles also, by biasing them more tightly into engagement with retainers 32 and 34, secures the poles in their axial positions with respect to the kayak body. In one or more other embodiments omitting tethers 36, in which retainer 32 is made of a polymer material, the downward-facing flanges are flexible and have inwardly (toward each other) facing protrusions on the flange ends, and the slot between the flanges is of a width slightly smaller than the pole shaft's diameter, the slot flanges receive the pole shaft in a releasable snap fit.

[0046] Referring to FIGS. 2, 3, 4, and 5, each of the two retainers 34 on a kayak body side 18 or 20 is secured to the kayak by a pair of T-bolts 50 extending upward from a universal track rail 52 secured to a respective body side edge 18 or 20 generally midway on the body, adjacent chair 24 and the operator's sitting position therein. The T-bolt shafts extend upward through respective slots 54 in an elongated aluminum bracket 56 of retainer 34. A respective threaded nut 58 is threaded down onto the upward-facing distal end of each T-bolt shaft and is tightened down onto the bolt thread to engage the top of bracket 56 and thereby secure retainer 34 to the kayak body. Bracket 56 includes an aluminum end plate 60 that is attached to the elongated bracket portion (herein "arm") by a pair of Torx-headed screws 62 (FIG. 5), that depends downward from the arm, and that adjacently faces an aluminum retainer member 64 that defines a through-bore 66 in which stake-out pole shaft 38 is slidably received.

[0047] Aluminum end plate 60 defines a forward face 68 that faces a rearward face 70 of retainer member 64. A generally cylindrical boss 72 extends from forward face 68 and is received in a through bore 74 of retainer member 64 so that retainer member 64 is rotatable about the axis of boss 72. In one or more embodiments, through bore 74 is countersunk into face 68, and the outer diameter of boss 72 is about 0.005" less than the diameter of the initial countersunk portion (88) of through bore 74, thereby allowing relative rotation between those components. When retainer member 64 and end plate 60 are so assembled together that boss 72 is received in bore 74 (for example, as in FIGS. 3 and 4), a generally cylindrical pin 76 extending forward from face 68 extends into an arcuate groove 78 extending into rearward face 70.

[0048] Face 70 of retainer member 64 appears larger than face 68 of end plate 60 in FIG. 5 because retainer member 64 has greater depth than does end plate 60, but it should be understood that the outer perimeter dimensions of faces 68 and 70 are generally equal. With end plate 60 and retainer member 64 abutting each other but otherwise aligned as they

are shown in FIGS. 3 and 4 (also shown in FIG. 5, except that retainer member 64 is rotated 180° about the righthand (in the perspective of FIG. 5) edge of end plate 60), pin 76 is received at the lefthand end (in the perspective of FIG. 5) end of arcuate groove 78 so that the end wall 80 of groove 78 blocks the retainer member's further rotation about boss 72, in the rotational direction of arrow 86. Pin 76 and groove edge 80 thereby define a stop between retainer member 64 and bracket 56 that blocks the retainer member's rotation with respect to the bracket in rotational direction 86, beyond the retainer member's orientation with respect to the bracket as shown in FIGS. 3 and 4 (in the direction away from the retainer member's orientation with respect to the bracket when pin 76 abuts the opposing edge 84 of groove 78, which is illustrated in FIG. 6).

[0049] With end plate 60 and retainer member 64 abutting each other but otherwise aligned as they are shown in FIG. 6, pin 76 is received at the righthand end (in the perspective of FIG. 5) of arcuate groove 78 so that end wall 84 of groove 78 blocks the retainer member's further rotation about boss 72, in the rotational direction of arrow 82. Pin 76 and groove edge 84 thereby define a stop between retainer member 64 and bracket 56 that blocks the retainer member's rotation with respect to the bracket beyond the retainer member's orientation with respect to the bracket shown in FIG. 6 (in the direction away from the retainer member's orientation with respect to the bracket when pin 76 abuts the opposing edge 80 of groove 78, which is illustrated in FIGS. 3 and 4). Arcuate groove 78 is configured so that, considering a plane that includes the center axis of through bore 74 and the center axis of pin 76 when it abuts end 80 of groove 78 and a plane that includes the center axis of through bore 74 and the center axis of pin 76 when it abuts end 84 of groove 78, those planes intersect, in one or more embodiments, at an angle of generally 90°, and so that an arc half way between the inner and outer surfaces of groove 78 is generally at a constant radius centered at the central axis of through bore 74. The radial width of groove 78, extending between the groove's inner and outer surfaces, is slightly larger than the diameter of pin 76 in the same direction, thereby facilitating ease of relative movement between the pin and the groove.

[0050] Through bore 74 is countersunk, with a larger diameter non-threaded portion 88 extending into rearward face 70 of retainer member 64 and a smaller diameter secondary portion 90 defined by the countersunk ledge in the middle of the through bore. Through bore 74 is also countersunk extending inward from the retainer member's forward face 92, so that the countersunk shelf that defines bore portion 90 is relatively narrow. Boss 72 is received by larger diameter portion 88 of bore 74 so that the boss's central through bore 94 is coaxially aligned with smaller diameter portion 90 of bore 74. As noted, the outer diameter of boss 72 is smaller than the diameter of larger diameter bore portion 88 by about 0.005", but it is larger than the diameter of the central, smaller bore, portion 90. The axial length of boss 72 is shorter than the depth of larger diameter portion 88, so that the distal end of boss 72 does not engage the countersunk flange in bore 74 that forms smaller diameter portion 90. A generally cylindrical bushing (not shown) is disposed against the distal end of boss 72 so that a non-threaded through bore of the bushing (which has the same diameter as the boss's threaded bore 94) is coaxial with the boss's threaded bore 94. The bushing's outer diameter is smaller than the diameter of central bore portion 90, so that

the central flange that defines bore portion 90 permits retainer member 64 to rotate easily about the bushing. The bushing's axial length is greater than the difference between the axial length of boss 72 and the axial depth of bore portion 88 by about 0.010", so that, when retainer member 64 is mounted onto end plate 60 about boss 72, the bushing extends through central portion 90 into the larger diameter portion through bore 74 that opens into forward surface 92. A 3/8-inch diameter hex head bolt (not shown) is inserted into the central bore of the bushing from the forward side 92 of retainer member 64 and the larger-diameter portion of bore 74 that opens into forward side 92, so that the bolt's 16 thread/inch threaded distal end engages the corresponding female thread of bore 94 of boss 72. In an assembled state, the bolt is threaded into bore 94, and the bolt's hex head remains forward of bore 90 but within the larger diameter bore portion that opens into the retainer member's forward face 92. The hex head extends radially outward of the perimeter of central bore portion 90, so that, when the bolt is tightened into bore 94, the bolt head retains the retainer member from moving away from bracket end plate 60 along the axis of through bore 94. The approximately 0.010" difference between the axial length of boss 72 and the axial depth of bore portion 88 permits an approximately 0.010" play between the position of retainer member 64 along the axis of bore 94 at which face 68 engages face 70 and the position of retainer member 64 at which the flange that defines central bore portion 90 engages the bolt head, thereby avoiding a frictional lock between faces 68 and 70 when the bolt is tightened onto the bushing and into the boss, facilitating ease of relative rotation between retainer member 64 and end plate 60.

[0051] The head of the hex bolt is received by a hexagonal opening in a base end 96 of a crank arm 98, that extends into the countersunk portion of through bore 74 that opens into forward face 92 so that manual actuation of crank arm 98 to rotate the crank arm around the axis of base end 96, which is coaxial with the axis of the bolt, rotates the bolt. Crank base end 96 has an end portion that engages the shoulder that defines central bore portion 90, a portion of face 92 that surrounds bore 74, or both. Depending on the direction of that rotation, the crank arm either tightens the bolt further into, or backs the bolt out of, through bore 94. Upon the operator's actuation of crank arm 98 to tighten the bolt into bore 94, the crank arm base biases rearward face 70 of retainer member 64 into forward face 68 of bracket end plate 60, thereby increasing the friction between the two components to a level sufficient to resist the torque on the retainer member caused, when the retainer member is in the position as shown in FIGS. 3 and 4, by the weight of stake-out pole 30 so that the retainer member remains in its orientation with respect to bracket 56 and to resist the torque on the retainer member caused, when the retainer member is in the position as shown in FIG. 6 and the operator has pushed the stake-out pole into the bed of the water body, by wind or waves acting upon the kayak so that the retainer member remains in its orientation with respect to bracket 56 as shown in FIG. 6. Upon the operator's actuation of crank arm 98 to back the bolt in the outward direction in bore 94, the crank base loosens that bias, or even slightly disengages from the retainer member, thereby releasing the frictional lock between faces 68 and 70 and allowing the axial play created by the bushing length to allow the retainer member to rotate freely about boss 72 and the axis of bore 94.

[0052] As indicated above, retainer member 64 forms a collar about shaft 38 of stake-out pole 30, retaining the stake-out pole shaft in position with respect to the retainer member. In one or more embodiments, the collar is formed by the complete encirclement of a perimeter of the stake-out pole shaft by the material of retainer member 64, forming through bore 66, as shown in FIG. 4, though it should be understood in view of the present disclosure that other configurations of the collar are possible, for example comprising a snap release bracket in an enclosure by the retainer member that is less than 360°. Thus, it should be understood that the example of the retainer member, and other structures, illustrated in the embodiments discussed herein are for purposes of example only.

[0053] The construction of the opposing faces 68 and 70 of bracket end plate 60 and retainer member 64 of the starboard side retainer 64 are the same as those of the port side retainer 34, as shown in FIG. 5, except that groove 78 is the mirror image of the groove 78 of the port side retainer about a vertical (in the perspective of FIG. 5) plane bisecting face 68 of bracket end plate 60. If the starboard side faces 68 and 70 are viewed in the orientation as shown in FIG. 5, edge 80 of groove 78 remains directly below through bore 74, but groove 78 extends to the left (in the view of FIG. 5) rather than to the right, and groove edge 84 is to the left of through bore 74. This change, then, reverses rotational directions 82 and 86, so that the operator deploys retainer member 64 from the transport position to the anchoring position by rotating retainer member 64 toward the bow. In one or more other embodiments, the retainer member and bracket end plates on the port and starboard sides have the same configurations. In such embodiments, the orientation of starboard-side retainer 32 is reversed, so that the slot defined by its end flanges opens upward. The starboard-side stake-out pole is reversed, so that the pole's handle is at the pole's forward end. Retainer member 64 rotates toward the kayak's stern in moving retainer 64 to the anchoring position, rotating the pole's forward end upward, out of retainer 32.

[0054] Referring to FIGS. 3, 4, and 6-8, the diameter of collar through bore 66 is greater than the outer diameter of shaft 38 of stake-out pole 30. As discussed above, the force applied to the stake-out pole by tether 36 inhibits the stake-out pole's movement axially with respect to the through bore's axis. Thus, in one or more embodiments, no further retention mechanism is included in the system to inhibit the pole's axial movement with respect to retainers 32 and 34. In one or more other embodiments, however, a selective lock is implemented, in addition to or instead of tether 36, to retain the pole axially with respect to the through bore's axis by a manually operable polymer (or other material) screw 100 that is threaded through a correspondingly threaded through bore through a top surface 102 of retainer member 64, so that a distal end of the polymer screw engages the outer diameter of pole shaft 38 and pushes the opposite side of the pole shaft against the far side of through bore 66, frictionally retaining the pole shaft with the screw's distal end and with the opposing side of through bore 66. It should be understood that this arrangement is presented for purposes of example only and that other arrangements, such as clamps, grommets or other suitable structures held by a retainer member and engaging the pole shaft, may be used to resist the pole's axial movement with

respect to the retainer member. Thus, the illustrated example is not a limitation of the present disclosure.

[0055] Referring to FIGS. 1, 2, 3, 5, and 6, the operator controls the orientation of retainer member 64 with respect to bracket 56 to move the two stake-out poles 30 between transport positions and anchoring positions. For each pole, when its retainer member 64 is in the position with respect to bracket 56 at which pin 76 abuts end 80 of groove 78 in face 68 of bracket end plate 60, retainer member 64 is oriented, as shown in FIGS. 3 and 4, so that the stake-out pole is in a transport position in which the axis of through bore 66 is aligned generally horizontally (assuming the disposition of kayak 10 on a body of water), so that the stake-out pole's dimension of elongation is generally parallel to the kayak's dimension of elongation. The pole is disposed outboard of and generally parallel to the hull's side, just beneath the deck's edge (or at least beneath a highest point of the deck), where the deck meets the hull. Because the bracket extends outward of a perimeter of the kayak body defined by the bow, the stern, and opposing body sides 18 and 20 so that the retainer member is outward of the perimeter, yet is also close to the hull and not above the level of the deck edge, the pole's position minimizes the likelihood that the pole will interfere with the operator's reach when paddling the kayak or operating accessory equipment. The pole's forward end is received by the slot in retainer 32 and is biased into the inward flange thereof by tether 36, thereby minimizing the likelihood that the pole will undesirably flex into the hull at the kayak's forward end. With the stake-out poles in this position, with the retainer members 64 tightened down into the position as shown in FIGS. 3 and 4 by crank handles 98, and with the poles secured in retainer 34 by screws 100 and/or tether 36, the operator launches the kayak into the body of water and drives the kayak, e.g. through paddling or a trolling motor, to a desired location, e.g. at which the operator wishes to fish.

[0056] If, on the water, the operator desires to anchor the kayak, the operator disengages tether 36 from the carabiner (or releases the carabiner from its attachment to the kayak body), thereby releasing one of the stake-out poles from that restriction. The operator reaches down to the pole's retainer 34 and grips and turns handle 98 to thereby loosen the corresponding retainer member 64 from its frictional engagement with the bracket end plate 60. Assuming the retainer 34 is on the kayak's port side, the operator then grips retainer member 64 and rotates it with respect to bracket end plate 60 about boss 72 in direction 82, toward the kayak bow. As retainer member 64 starts rotating, the stake-out pole's engagement in the retainer member causes the stake-out pole to rotate in the same direction, causing the shaft 38's distal end to move downward, out of engagement with its retainer 32. If the operator continues to rotate the retainer member, pin 76 eventually engages end 84 of groove 78, at which point retainer member 64, and stake-out pole 30, have rotated approximately 90°, to the position shown at FIG. 6, so that the stake-out pole's dimension of elongation is generally perpendicular to the kayak's dimension of elongation. The operator manually turns screw 100, if present, in the direction to back the screw out of its threaded hole through the retainer member, thereby releasing the screw's engagement with the pole shaft. The operator then grips stake-out pole handle 40 and pushes the pole downward and

into the water body bed, thereby anchoring the kayak. The operator may repeat this operation with the stake-out pole on the kayak's starboard side.

[0057] Because neither tether 36 nor screw 100 now engages the pole shaft in either of the port and starboard retainers, and each retainer member can therefore move with respect to its pole shaft, the kayak can freely move up and down with waves or tide changes.

[0058] The process reverses when the operator wishes to move the kayak. The operator pulls up on handle 40 of the port side stake-out pole until it clears the water body bed and the pole reaches the position at which the operator wishes to secure the pole in retainer member 64. The operator actuates screw 100, if present, in the direction into the threaded hole in the top of the retainer member until the screw's distal end engages the pole shaft to thereby frictionally retain the pole shaft (in its axial direction) between the screw's distal end and the opposing side of through bore 66. The operator then grips retainer member 64 and rotates it, in direction 86, from the position shown in FIG. 6 back to the position shown in FIGS. 2 and 3, at which point pin 76 again engages end 80 of groove 78 and the end of pole shaft 38 reengages with the yoke slot of retainer 32. The operator wraps tether 36 back around pole shaft 38 and retains the tether with the carabiner. The operator repeats the process with the starboard-side stake-out pole (if it had been deployed) and thereafter moves the kayak to the next desired location.

[0059] FIGS. 9 and 10 illustrate a still further embodiment of retainer member 64 and bracket plate 60 in which a spring-loaded plunger, or pop pin, 104 replaces both the hand-actuatable crank 96/98 (FIGS. 3, 6, 7, and 8) and the pin 76 (FIG. 5) of the above-described embodiments. In embodiments as in FIGS. 9 and 10, aluminum end plate 60 defines a forward face 68 that faces a rearward face 70 of retainer member 64. A generally cylindrical boss 72 extends from forward face 68 and is received in a double countersunk through bore 74 of retainer member 64. The diameter of boss 72 is slightly less (in one or more embodiments, for example, approximately 0.005" less) than the diameter of the large diameter portion 88 of the countersunk bore of bore 74 extending into retainer member face 70, thereby permitting retainer member 64 to rotate about the axis of boss 72. A cylindrical bushing (not shown) having an outer diameter slightly less than the diameter of a smaller-diameter central portion 90 of through bore 74 defined by a relatively narrow shoulder between the two larger-diameter countersunk portions of bore 74 has an axial length that is approximately 0.010" longer than the difference in axial length between boss 72 and initial countersunk bore portion 88 of bore 74, so that the bushing extends through smaller-diameter central portion 90 as it extends from the distal end of boss 72. The bushing has a non-threaded through bore having approximately the same diameter as the diameter of threaded bore 94 centered in boss 74, and the two bores are coaxially aligned with each other. A threaded bolt (not shown) extends into bore 74 from the countersunk portion 118 of bore 74 that opens into forward surface 92 of the retainer member so that the bolt's threaded distal end is threadedly received in threaded bore 94 of boss 72. The bolt's head remains on the opposite side of the inwardly depending shoulder in bore 74 that defines smaller-diameter portion 90, within the countersunk portion of bore 74 that opens into forward face 92. The bolt head's diameter is greater than the diameter of center bore portion 90, thereby retaining retainer member 64

against movement away from end plate 60 along the axis of bores 74 and 94. The bushing is about 0.010" longer than the difference between the lengths of boss 72 and countersunk bore portion 88 (boss 72 is shorter than the depth of bore portion 88) so that retainer member 64 can move slightly in either direction along the axial dimension of bores 94 and 74 (between a position of retainer member 64 when it engages the opposing surface of the bracket end plate and a position of retainer member 64 when the inwardly depending shoulder that defines central bore portion 90 engages the bolt head), even when the bolt is fully threaded into bore 94 of boss 72. This facilitates relative rotation between the retainer member and the bracket end plate, in that the bolt's engagement into threaded bore 94 does not frictionally lock retainer member face 70 against bracket end plate face 68. When retainer member 64 and end plate 60 are assembled together (bracket end plate 60 is attached to the bracket arm, which is attached to a universal track rail in the same manner as discussed above, by a pair of screws that extend into the countersunk bores at the top of the bracket end plate as shown in FIG. 10) so that boss 72 is received in bore 74 and the retainer member and bracket end plate are dimensionally aligned with each other (for example, similar to the arrangement as in FIGS. 3, 4, and 9), a generally cylindrical pin (not shown) of spring-loaded plunger 104 extends through bracket member rearward face 70, from a through hole 106, and extends from rearward face 70 into an arcuate groove 78 defined in forward face 68 of the bracket end plate.

[0060] Spring-loaded plungers are well understood, and their specific construction and operation is, therefore, not discussed in detail herein as it applies generally with regard to the variety of forms and operations in which they may be provided. In one or more examples as illustrated in FIGS. 9 and 10, however, spring-loaded plunger 104 has a handle end 110 with a hollow shaft portion 112 that receives an elongated pin (not shown) in a fixed relationship. This plunger pin passes through a central through bore of an annular collar 114 that is separate from handle 110 and shaft portion 112 and that has a threaded lower end (not shown) that threads into a corresponding female thread at the end of bore 106 as that bore opens into forward face 92 of retainer member 64, thereby fixing the spring-loaded plunger in the retainer member. At a lower portion of the plunger pin, but short of the pin's distal end, there is a radially extending annular shoulder (not shown) having an outer diameter that is slightly smaller than the inner diameter of bore 106 so that the pin, including the annular collar, moves freely in bore 106 in the bore's two axial directions but generally does not move from side to side. A coil spring (not shown) abuts, at one end of the spring, the top of the pin's annular shoulder and, at the opposite end of the spring, the bottom of threaded annular collar 114 (e.g., either at the bottom of the threaded portion itself or at a countersunk inner shoulder within the threaded portion). The spring is in compression, so that it thereby biases the pin toward bracket end plate 60 and into annular groove 78 and, therefore, biases handle 110/112 into engagement with the top of annular collar 114, in the position shown in FIG. 9. If the operator, however, grips handle 110 and pulls the handle away from threaded annular collar 114 against the spring bias, the handle moves away from collar 114, pulling the pin through collar 114 and partially out of bore 106 through face 92 of retainer member 64, further compressing the spring. If the operator releases handle 110, the spring moves the pin and handle back into

the position shown in FIG. 9. In view of the present disclosure, it will be apparent that variations in the structure and operation of the spring-loaded plunger may be employed and that the present discussion is for purposes of example only.

[0061] Face 70 of retainer member 64 appears larger than face 68 of end plate 60 in FIG. 10 because retainer member 64 has greater depth than does end plate 60, but it should be understood that the outer perimeter dimensions of faces 68 and 70 are generally the same. FIGS. 9 and 10 illustrate a retainer member and bracket end plate from a starboard-side retainer 64. With end plate 60 and retainer member 64 abutting each other but otherwise aligned as they are shown in FIG. 9, and similar to the arrangement in FIGS. 3 and 4 (also similar to the arrangement shown in FIG. 5, except that retainer member 64 of FIG. 10 is rotated 180° about an axis parallel to the lefthand (in the perspective of FIG. 10) edge of end plate 60), the plunger pin is received at the righthand end (in the perspective of FIG. 10) end of arcuate groove 78 in a blind bore 108 in groove 78 so that the pin's engagement within bore 108 blocks the retainer member's further rotation about boss 72, in the directions both of arrow 82 and arrow 86. The plunger pin and bore 108/groove edge 80 thereby define a stop between retainer member 64 and bracket 56 (see FIG. 3) that blocks the retainer member's further rotation with respect to the bracket in direction 86, beyond the retainer member's orientation with respect to the bracket as shown in FIG. 9 in the direction away from the retainer member's position when the pin is at the other end of groove 78, while bore 108 also blocks the retainer member's rotation in the opposite rotational direction (82).

[0062] In one or more embodiments, spring-loaded plunger 104 is chosen of a size so that, even when the operator pulls handle 110 to its maximum retracted position, so that the plunger spring is at its position of maximum compression, the plunger pin's distal end retracts from bore 108 but does not retract fully into bore 106. Thus, in handle 110's retracted position, the pin's distal end is retracted clear of bore 108 but remains inserted into arcuate groove 78. Accordingly, when the handle/plunger pin is in this maximum retracted position, the pin's distal end can move in either direction within arcuate groove 78 but not beyond the groove's end walls 80 and 84. If the operator releases the handle when the pin is over bore 108, or over a bore 116 at the opposite end of arcuate groove 78, such that the spring biases the pin's distal end into either bore, the pin, and therefore retainer member 64 via the pin shaft's location within bore 106, is rotationally fixed with respect to bracket end plate 60. If, on the other hand, the operator releases the handle when the pin is over arcuate groove 78 between bores 108 and 116, the plunger pin's tip engages but moves within the trough of groove 78, and the retainer member can be rotated in either direction 82 or 86 until the pin reaches either bore 108 or bore 116, at which point the spring biases the pin's distal end into the bore, thereby locking the retainer member rotationally with respect to end plate 60. To remove the lock (with respect to the arc of arcuate groove 78 but not beyond, due to end walls 80 and 84), the operator pulls back on handle 110 until the pin's distal end clears the bore 108 or 116 in which it happens to be received, thereby allowing the operator to rotate retainer member 64 in the one of rotational direction 82 or 86 that moves the pin's distal end in groove 78 toward the other bore 108 or 116.

[0063] With end plate 60 and retainer member 64 abutting each other but otherwise aligned so that the axes of through bore 66 (which is the same as, and is shown in, FIG. 4 of the embodiments discussed above, although on the starboard side rather than the port side) and the stake-out pole shaft are aligned vertically (as the retainer member and end plate are aligned as shown in FIG. 6, although again on the starboard side rather than the port side, with retainer member 64 having been rotated toward the bow), the plunger pin distal end is received at the lefthand end (in the perspective of FIG. 10) of arcuate groove 78 in a blind bore 116 in groove 78 so that the pin's engagement within bore 116 blocks the retainer member's further rotation about boss 72, in the directions both of arrow 82 and arrow 86. The plunger pin and bore 116/groove edge 84 thereby define a stop between retainer member 64 and bracket 56 (see FIG. 3) that blocks the retainer member's further rotation with respect to the bracket in direction 82, beyond the retainer member's orientation with respect to the bracket similar to that as shown in FIG. 6 in the direction away from the retainer member's orientation with respect to the bracket when the plunger pin abuts the opposing edge 80 of groove 78 (which is illustrated in FIG. 9), while bore 116 also blocks the retainer member's rotation in the opposite rotational direction (86). Arcuate groove 78 is configured so that, considering a plane that includes the center axis of through bore 74 and the center axis of bore 108 of groove 78 and a plane that includes the center axis of through bore 74 and the center axis of bore 116 of groove 78, those planes intersect, in one or more embodiments, at an angle of generally 90°, and so that an arc half way between the inner and outer surfaces of groove 78 is generally at a constant radius centered at the central axis of through bore 74. The radial width of groove 78, extending between the groove's inner and outer surfaces, is slightly larger than the diameter of the plunger pin in the same direction, thereby facilitating ease of relative movement between the pin and the groove.

[0064] The construction of the opposing faces 68 and 70 of bracket end plate 60 and retainer member 64 of the port side retainer 64 are the same as those of the starboard side retainer 34, as shown in FIG. 10, except that groove 78 is the mirror image of the groove 78 of the starboard side retainer about a vertical (in the perspective of FIG. 10) center plane bisecting face 68 of the bracket end plate and except that the position of through bore 106 through the retainer member is mirror image of the through bore's position of the starboard side retainer about a vertical (in the perspective of FIG. 10) center plane bisecting face 70 of the retainer member. If the port side faces 68 and 70 are viewed in the orientation as shown in FIG. 10, bore 106 in face 70 is below and to the right of through bore 74, rather than below and to the left, as in FIG. 10. Groove 78 remains directly below through bore 74, but the positions of edges 80 and 84, and of bores 108 and 116, are reversed. These changes, then, reverse rotational directions 82 and 86, so that the operator deploys retainer member 64 from the transport position to the anchoring position by rotating retainer member 64 toward the bow. In one or more other embodiments, the retainer member and bracket end plates on the port and starboard sides have the same configurations. In such embodiments, the port stake-out pole on extends from retainer 34 to a retainer 32 mounted on the kayak's stern, and retainer member 64 rotates toward the kayak's stern in moving the retainer to the anchoring position.

[0065] As discussed with respect to one or more embodiments above, the diameter of the collar through bore in the retainer member that receives the stake-out pole is greater than the outer diameter of shaft 38 of stake-out pole 30 (FIGS. 1 and 2, and similar to FIG. 8). Although not present in the embodiment of FIGS. 9 and 10, in one or more similar embodiments, to retain the pole axially with respect to the through bore's axis, a manually operable polymer screw, such as a screw 100 as in FIG. 8, is threaded through a correspondingly threaded through bore through a top surface 102 of retainer member 64, so that a distal end of the polymer screw engages the outer diameter of the pole shaft, thereby establishing a frictional lock between the screw and the pole shaft and between the pole shaft and the opposing inner surface of the through bore.

[0066] Referring additionally to FIGS. 1, 2, 3, 5, and 6, the operator's control of the orientation of the embodiments of retainer member 64 as illustrated in FIGS. 9 and 10 with respect to bracket 56 to move the two stake-out poles 30 between transport positions and anchoring positions is similar to that discussed above with respect to FIGS. 1-8. For the starboard side pole, when its retainer member 64 is in the position with respect to bracket 56 at which the plunger pin is received in blind bore 108 and abuts end 80 of groove 78 in face 68 of bracket end plate 60, retainer member 64 is oriented, similar to the orientation as shown in FIGS. 3 and 4, so that the axis of through bore 66 is aligned horizontally and the stake-out pole is in the transport position, as discussed above. With both stake-out poles in this position (and secured by the retainers 32 and tethers 36), with each plunger pin received in its blind bore 108 to thereby lock the retainer members with respect to the brackets, the operator may launch the kayak into the body of water and drive the kayak, e.g. through paddling or a trolling motor, to a desired location.

[0067] To anchor the kayak, and assuming actuation of the starboard side pole, the operator disengages starboard tether 36, as described above, and reaches down to the starboard retainer 34 and pulls handle 110 of spring-loaded plunger 104 outward, away from forward face 92 of retainer member 64 to thereby retract the plunger pin's distal end from blind bore 108, but retaining the pin's distal end within arcuate groove 78, thereby allowing the pin to move within the groove and the retainer member to rotate with respect to the bracket end plate to the extent permitted by the plunger pin's engagement with groove 78. The operator then rotates retainer member 64 with respect to bracket end plate 60 about lug 72 in direction 82. As retainer member 64 starts rotating, and the plunger pin's distal end clears the opening of bore 108, the operator may release plunger handle 110. The spring of spring-loaded plunger 104 then pushes the pin down into engagement with the bottom of arcuate groove 78 between bores 108 and 116, but the pin's tip slides along the groove with the retainer member's rotation. As the retainer member thus rotates, the stake-out pole's engagement in the retainer member causes the stake-out pole to rotate in the same direction, causing the shaft 38's distal end to move downward, out of engagement with its retainer 32. If the operator continues to rotate the retainer member, the plunger pin eventually engages end 84 of groove 78, at which point the plunger spring biases the plunger pin's distal end down into blind bore 116, again rotationally locking the retainer member with respect to the bracket end plate. At this point, retainer member 64, and stake-out pole 30, have rotated

approximately 90°, to an anchoring position such as that shown at FIG. 6 (although on the kayak's starboard side, rather than the port side). The operator then grips stake-out pole handle 40 and pushes the pole downward and into the water body bed, thereby anchoring the kayak. The operator may repeat this operation with the stake-out pole on the kayak's port side.

[0068] The process reverses when the operator wishes to move the kayak. The operator pulls up on handle 40 of one of the stake-out poles until it clears the water body bed and the pole reaches the position at which the operator wishes to secure the pole in retainer member 64. The operator then pulls plunger handle 110 outward, away from forward face 92 of retainer member 64 to thereby retract the plunger pin's distal end from blind bore 116, while retaining the pin's distal end within arcuate groove 78, thereby allowing the pin to move within the groove and the retainer member to rotate with respect to the bracket end plate to the extent permitted by the plunger pin's engagement with groove 78. The operator then rotates retainer member 64 with respect to bracket end plate 60 about lug 72 in direction 86, from the position such as shown in FIG. 6 back to the position such as shown in FIGS. 2, 3, and 9, at which point the plunger pin again engages end 80 of groove 78 and is biased by the plunger spring back into blind bore 108, and the end of pole shaft 38 reengages with the yoke slot of retainer 32. The operator wraps tether 36 back around pole shaft 38 and retains the tether with the carabiner. The operator repeats the process with the other stake-out pole (if it had been deployed) and thereafter moves the kayak to the next desired location.

[0069] Referring to FIGS. 11-13, a retainer 34 is used to secure to kayak 10 a watercraft accessory other than a stake-out pole. In this example, the accessory is a fish finder sonar transducer and an associated mounting pole therefor, but it should be understood that other accessories may be used. In the illustrated one or more embodiments, retainer 34 is constructed in either of the two embodiments discussed above, or in another suitable arrangement in view of the present disclosure, except that the cross sectional diameter of through bore 66 through which a transducer mounting pole 120 extends is increased to approximately 1.51" in order to accommodate an approximately 1.5" cross-sectional diameter of the main shaft of pole 120 (over its entire length below its handle), as opposed to the diameter of stake-out pole shaft 38 (FIGS. 2, 6, and 8). In other embodiments, the diameter of through bore 66 may be larger, to the extent greater play between the two components is desired. As the construction and operation of retainer 34, and the manner of its mounting to kayak 10, in these embodiments are the same as discussed above with respect to the embodiments discussed above with respect to FIGS. 1-10, that discussion is not repeated but should be understood to be incorporated herein.

[0070] Transducer mounting pole 120 includes a mounting base for a sonar transducer, in this example an elongated main shaft 122 that has a hollow interior, thereby defining an interior volume about an axis 124 in the dimension of the mounting pole's elongation. In one or more embodiments, main shaft 122 is made of fiberglass but, in other embodiments, may be made of aluminum or other suitable material. A lower through hole 126 extends through the wall of mounting pole 120 proximate a lower end 128 of the pole, while an upper through hole 129 extends through the mount-

ing pole's wall proximate an upper pole end 130. Due to the pole's hollow construction, the two through holes are in operative communication with each other.

[0071] In one or more embodiments, a bracket 132 connected to a sonar transducer 134 is releasably attached to mounting pole 120 proximate pole end 128. One or more communication and/or power electrical wires or cables 138 extend from transducer 134, through the hollow volume of main shaft 122, and out of hole 129. Electrical cables 138 extend to and electrically connect to a processing unit (not shown) that houses the fish finding system's data collection and processing capability. The processing unit is not accessed by the kayak operator during use and so is disposed in an out-of-the-way location in the watercraft, e.g., under the operator's chair. A further cable 138 extends from the processing unit to a display monitor 160, as illustrated in FIGS. 12 and 13, that receives signals via the cables that correspond to the data acquired by sonar transducer 134 and that drive display monitor 160 to display images of objects below the deck, and in one or more embodiments below the hull, of kayak 10, such as fish, the waterway bottom surface, and debris and other waterway bottom features. For ease of illustration, the processing unit is not illustrated in the Figures, and cables 138 are illustrated as running directly from the transducer to display unit 160, but it should be understood that the present disclosure encompasses the inclusion of the intermediate processing unit.

[0072] Main shaft 122 passes through an annular collar 146 that surrounds the main shaft. A set screw 148 passes through the wall of annular collar 146 and is threadedly engaged with the hole through the annular collar wall. Thus, set screw 148 may be turned in either rotational direction to extend further into, or retract away from, the collar's central opening. By tightening the screw down onto the outer surface of tube main shaft 122, so that the set screw's threaded end engages the main shaft, when the collar is located at a desired position on the main shaft, the operator secures the collar at the desired position. Because the outer surface dimensions of collar 146 are greater than those of the circumference of through bore 66 through retainer member 64, collar 146 provides a stop to define the main shaft's maximum sliding position with respect to retainer 34 (and, thereby, with respect to the kayak hull), in one direction along the axis of through bore 66. Collar 146 thereby acts as a fast location device by which the operator can insert distal end 128 of the main shaft into and through through-bore 66 and slide the main shaft to the desired position, e.g., a position such that when the transducer mounting pole is disposed by retainer 34 in the deployed position, distal end 128 is sufficiently below the waterline and below, or otherwise oriented with respect to, the kayak hull that the kayak hull does not interfere with the sonar signal of sonar transducer 134 such as indicated in FIG. 13. Collar 146 thus acts as a height gauge for the main shaft that thereby easily locates the sonar transducer at a desired depth underwater with respect to the kayak deck and/or hull. By loosening set screw 148, and thereby releasing the collar on the main shaft, the operator can alter the collar's position on the main shaft as desired, tightening the screw at the new position, or may remove the collar altogether.

[0073] Opposing through holes 140 and 142 through the wall of main shaft 122 receive an approximately 0.75" fiberglass (or other suitable material, such as aluminum) handle shaft 144 so that handle shaft 144 extends entirely

through main shaft 122 proximate its upper end 130. Holes 140 and 142 are slightly larger in cross section than the outer diameter of handle shaft 144 so that handle shaft 144 is slidably received therethrough. A handle grip 158 at one end of handle shaft 144 has a diameter greater than that of through holes 140 and 142, thereby precluding the handle from falling through the holes and into the water body when the transducer mounting pole is in the transport position, as shown in FIG. 12.

[0074] A steel end cap 150 of handle shaft 144 opposite collar 146 with respect to the tube's main shaft 122 is pointed, both facilitating the handle shaft's insertion through the main shaft holes 140 and 142 and, depending on the orientation of transducer 134 on the tube at the tube's lower end, providing a visual indication of that orientation to the kayak operator. Generally, the transducer is attached to the tube main shaft so that, when the transducer is mounted on the main shaft, the transducer's sonar beam is directed downward. As should be understood, however, sonar transducer brackets (see 132) are known that permit the transducer to have variable orientation with respect to the collar, such that once the transducer is mounted onto pole main shaft 122, e.g. when retainer 34 holds main shaft 122 in the position as shown in FIG. 13, the operator can adjust the orientation of the transducer with respect to the main shaft so that, when retainer 34 holds main shaft 122 in the deployed position as shown in FIG. 12, the transducer's sonar beam is directed in a desired direction, e.g., generally horizontally in order to detect fish in covered locations. Tube holes 126 and 140 are in the same rotational position on the tube main shaft. If the operator, when mounting the transducer on the tube main shaft's lower end, secures the transducer by bracket 132 so that the transducer is rotationally aligned on the main shaft on the opposite side of the main shaft from hole 126, and if the operator inserts the handle shaft initially through hole 140 so that pointed end 150 is on the side of the tube main shaft of hole 142, pointed end 150 indicates the transducer's location with respect to the tube main shaft. If the operator also knows the transducer's orientation with respect to main shaft 122, pointed end cap 150 also provides information about the direction of the transducer's sonar beam. Further, by loosening set screw 100 (FIGS. 7 and 8) in retainer 34 while retainer 34 is in the deployed position as shown in FIG. 12 and main shaft 122 is vertically secured because collar 146 is secured to main shaft 122 and rests on retainer 34, the operator can rotate the tube main shaft (through manipulation of handle shaft 144) within retainer through bore 66 (FIGS. 7 and 8) about the main shaft's axis of elongation until handle shaft pointed end 150 points in a direction at which the operator desires to locate the sonar transducer, thereby locating the sonar transducer (and, depending on the transducer's orientation with respect to main shaft 122) in that desired position.

[0075] A clear anodized aluminum end cap 152 is disposed at upper end pole end 130. A set screw 154 passes through the wall of end cap 152 and is threadedly engaged with the hole through the end cap. Thus, set screw 152 may be turned in either rotational direction to extend further into, or retract away from, the volume beneath the cap and, because end 130 of tube main shaft 122 at end 130 along axis 124 is open, within the interior of tube main shaft 122. Set screw 154 is of a length sufficient to reach the outer surface of handle shaft 144 when the set screw is disposed in the threaded hole through cap 152. Thus, by tightening the

screw down onto the outer surface of handle shaft **144**, so that the set screw's threaded end engages the handle shaft, the operator secures the handle shaft in position with respect to tube main shaft **122**.

[0076] As apparent in view of the discussion immediately above in view of the embodiments described in conjunction with FIGS. 1-10, retainer **34** secures and maintains tube main shaft **122** in position with respect to the kayak hull, selectively in a transport position in which the elongated main shaft is disposed horizontally adjacent to the hull and proximate the deck edge and in a deployed position in which the main shaft is disposed vertically adjacent to the hull and the deck edge so that the sonar transducer is below the hull's waterline a distance at least sufficient so that the kayak hull does not interfere with the transducer's sonar beam (this distance may be adjusted by the operator, e.g. through manipulation of collar **146**, to accommodate selected positioning of the transducer with respect to main shaft **122**, as discussed above). Thus, in one or more embodiments, the pole is movable by the operator between the transport and deployed positions by an approximately 90° rotational movement of retainer **34**. Generally, the operator maintains transducer mounting pole in the transport position that is parallel (in terms of its dimension of elongation) with the dimension of elongation of the kayak deck and so that tube main shaft and the sonar transducer are both above the water line, adjacent the hull when paddling or trolling between a water put in/take out position and a fishing area on the water body and in the deployed position when in the fishing area. In one or more arrangements, the operator may secure a first retainer **34** on one side of the kayak to hold the sonar transducer system and a second retainer **34** on the other side of the kayak to hold a stake-out pole, but, in other arrangements, the operator has only one retainer **34** to secure one or the other, or another, accessory. For instance, it is known, while fishing in a kayak in conjunction with fish finding sonar, to paddle or troll the kayak on the water body while monitoring the fish finder (see **160**), such that the sonar transducer and the stake-out pole are not deployed simultaneously.

[0077] With transducer mounting pole in the transport position, as discussed above and shown in FIG. 12, retainer **34** secures main shaft **122** so that it is aligned horizontally, adjacent the kayak hull. As the main shaft is shorter than the stake-out pole, no tether or secondary retainer (see **32**, in FIG. 2) is used to additionally secure the transducer mounting pole in position in the illustrated embodiment, but it should be understood that the present disclosure encompasses the use of a tether and/or a secondary retainer to additionally retain the transducer mounting pole in that position. To move the transducer mounting pole to the deployed position, and assuming that the pole is secured to the kayak by a retainer **34** on the kayak's port side (as reflected by the discussion above, a retainer **34** retaining transducer mounting pole **120** could be disposed on the kayak's starboard side) and, for purposes of explanation only, that an embodiment of retainer **34** as in FIGS. 9-10 is used, the operator disengages the tether, if present, that additionally secures the transducer mounting pole to the kayak, reaches down to the port retainer **34**, and pulls handle **110** of spring-loaded plunger **104** outward, away from forward face **92** of retainer member **64**. This retracts the plunger pin's distal end from blind bore **108**, but retains the pin's distal end within arcuate groove **78**, thereby allowing

retainer **34** to operate as discussed above with respect to FIGS. 9 and 10. In this example of a port-side retainer **34**, and referring again to FIGS. 9 and 10, the construction of the opposing faces **68** and **70** of bracket end plate **60** and retainer member **64** of the port side retainer **64** are the same as those of the starboard side retainer **34**, as shown in FIG. 10, except that groove **78** is the mirror image of the groove **78** of the starboard side retainer about a vertical (in the perspective of FIG. 10) center plane bisecting face **68** of the bracket end plate and except that the position of through bore **106** through the retainer member is mirror image of the through bore's position of the starboard side retainer about a vertical (in the perspective of FIG. 10) center plane bisecting face **70** of the retainer member. If the port side faces **68** and **70** are viewed in the orientation as shown in FIG. 10, bore **106** in face **70** is below and to the right of through bore **74**, rather than below and to the left, as in FIG. 10. Groove **78** remains directly below through bore **74**, but the positions of edges **80** and **84**, and of bores **108** and **116**, are reversed. These changes, then, reverse rotational directions **82** and **86**, so that the operator deploys retainer member **64** from the transport position to the deployed position by rotating retainer member **64** toward the bow. Thus, the operator then so rotates retainer member **64** with respect to bracket end plate **60** about lug **72** in direction **82** (again, reverse from the direction shown in FIG. 10). As the retainer member thus rotates, the transducer mounting pole's engagement in the retainer member causes the mounting pole to rotate in the same direction, causing the main shaft distal end **128** to move downward and, if a secondary retainer (such as retainer **32**, FIG. 2) is present, out of the secondary retainer. Referring also to FIG. 10 (although, as noted, in mirror image), if the operator continues to rotate the retainer member, the plunger pin eventually engages end **84** of groove **78**, at which point the plunger spring biases the plunger pin's distal end down into blind bore **116**, again rotationally locking the retainer member with respect to the bracket end plate. At this point, retainer member **64**, and transducer mounting pole **120**, have rotated approximately 90°, to an operative position such as that shown in FIG. 13. The distal end **128** of main shaft **122**, and sonar transducer **134**, now being below the water line in the deployed position, the operator connects signal and/or power cables **138** into the fish finder **160** for display of information provided by cables **138** from the transducer, actuates the display, and proceeds to operate the sonar system. As discussed above, the operator may adjust the sonar transducer's rotational position about axis **124** by loosening set screw **100** (FIG. 8) and manipulating handle shaft **144** until its pointing end **150** points in the desired direction for the sonar transducer's location, as discussed above.

[0078] The process reverses when the operator wishes to move the transducer mounting pole from the deployed position back to the transport position. Referring again also to FIGS. 9 and 10 (again, in mirror image), the operator pulls plunger handle **110** outward, away from forward face **92** of retainer member **64** to thereby retract the plunger pin's distal end from blind bore **116**, while retaining the pin's distal end within arcuate groove **78**, thereby allowing the pin to move within the groove and the retainer member to rotate with respect to the bracket end plate to the extent permitted by the plunger pin's engagement with groove **78**. The operator then rotates retainer member **64** with respect to bracket end plate **60** about lug **72** in direction **86**, from the position such as

shown in FIG. 13, back to the position such as shown in FIG. 12, at which point the plunger pin again engages end 80 of groove 78 and is biased by the plunger spring back into blind bore 108. If a retainer such as retainer 32 (FIG. 2) is present, main shaft 122 re-engages with the yoke slot of the secondary retainer. If a tether, such as tether 36 (FIG. 2) is used, the operator wraps tether 36 back around main shaft 122 and retains the tether with a suitable device, such as a carabiner.

[0079] Many modifications and other embodiments set forth herein will come to mind to one skilled in the art to which these embodiments pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. For example, while the examples discussed herein are described with respect to kayaks, it should be understood that the structures and methods as discussed herein may be used with other watercraft. Therefore, it is to be understood that the embodiments are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the invention. Moreover, although the foregoing descriptions and the associated drawings describe example embodiments in the context of certain example combinations of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the invention. In this regard, for example, different combinations of elements and/or functions than those explicitly described above are also contemplated within the scope of the invention. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

What is claimed is:

1. A retention system for use with a watercraft, the watercraft comprising an elongated body, comprising a forward end, a rearward end opposite the forward end along the body's dimension of elongation, a hull extending from the forward end to the rearward end and forming a bottom of the body, and a deck extending from the forward end to the rearward end and forming a top of the body, the retention system comprising:

- a bracket configured to be attached to the body;
- a retainer member attached to the bracket movably between a first orientation with respect to the bracket and a second orientation with respect to the bracket; and
- a mounting base received by the retainer member, the mounting base having a proximate end and an opposite distal end;

wherein, when the bracket is attached to the body and the retainer member is attached to the bracket,

in the retainer member's first orientation, the mounting base is received by the retainer member so that the mounting base distal end is adjacent the elongated body proximate the deck,

in the retainer member's second orientation, the mounting base is received by the retainer member so that the mounting base distal end is below the deck, and

in the retainer member's second orientation, the mounting base is outward of a side of the body extending between the forward end and the rearward end.

2. The retention system as in claim 1, further comprising a sonar transducer coupled with the mounting base distal end.

3. The retention system as in claim 1, wherein the watercraft is a kayak.

4. The retention system as in claim 1, wherein the bracket extends outward of a perimeter of the body defined by the forward end, the rearward end, and opposing sides of the body at one of the opposing sides of the body so that the retainer member is outward of the perimeter, wherein each opposing side extends between the forward end and the rearward end.

5. The retention system as in claim 1, wherein the retainer member comprises a collar that at least partially encloses an outer perimeter of the mounting base.

6. The retention system as in claim 5, comprising a selectively engageable lock between the collar and the mounting base.

7. The retention system as in claim 6, wherein the lock comprises a manually actuatable threaded screw that extends through a wall of the collar so that a distal end of the threaded screw is selectively engageable, upon the threaded screw's rotation, with an outer surface of the mounting base.

8. The retention system as in claim 1, wherein the retainer member is attached to the bracket pivotably about an axis.

9. The retention system as in claim 8, comprising a first stop between the retainer member and the bracket that blocks rotation of the retainer member with respect to the bracket beyond the first orientation away from the second orientation and a second stop between the retainer member and the bracket that blocks rotation of the retainer member with respect to the bracket beyond the second orientation away from the first orientation.

10. The retention system as in claim 8, comprising a lock between the bracket and the retainer member, the lock between the bracket and the retainer member being configured to resist movement of the retainer member out of the first orientation and to resist movement of the retainer member out of the second orientation.

11. The retention system as in claim 1, wherein the mounting base is an elongated, generally cylindrical member.

12. The retention system as in claim 11, comprising a handle attached to an end of the generally cylindrical member opposite the mounting base distal end.

13. The retention system as in claim 1, wherein

the bracket is configured to be attached to the deck and to extend across and outward of a perimeter of the body defined by the forward end, the rearward end, and opposing sides of the body at one of the sides of the body, wherein each opposing side extends between the forward end and the rearward end,

the bracket comprises a collar disposed at least partially outward of the perimeter;

the mounting base comprises an elongated pole received by the collar so that the collar at least partially encloses an outer perimeter of the pole,

wherein when the bracket is attached to the body and the collar is attached to the bracket,

in the collar's first orientation, the pole's dimension of elongation is generally parallel to the body's dimension of elongation, and

in the collar's second orientation, the pole's dimension of elongation is generally perpendicular to the body's dimension of elongation.

14. The retention system as in claim 12, further comprising

a selectively engageable first lock between the collar and the pole, and
 a selectively engageable second lock between the bracket and the collar, the second lock being configured to resist movement of the collar out of the first orientation and to resist movement of the collar out of the second orientation.

15. A retention system for use with a watercraft having an elongated body with a forward end, a rearward end opposite the forward end along the body's dimension of elongation, a hull extending from the forward end to the rearward end and forming a bottom of the body, and a deck extending from the forward end to the rearward end and forming a top of the body, the retention system comprising:

a bracket configured to be attached to the body;
 a retainer member attached to the bracket movably between a first orientation with respect to the bracket and a second orientation with respect to the bracket, wherein the retainer member is configured to receive an elongated pole; and

at least one watercraft accessory coupled with a distal end of the elongated pole,

wherein the retainer member is attached to the bracket so that, when the bracket is attached to the body and the elongated pole is received by the retainer member,

in the retainer member's first orientation, the elongated pole is received by the retainer member so that the elongated pole is adjacent the elongated body proximate the deck,

in the retainer member's second orientation, the elongated pole is received by the retainer member so that the elongated pole's dimension of elongation is generally perpendicular to the body's dimension of elongation, and

in the retainer member's second orientation, the mounting base is outward of a side of the body extending between the forward end and the rearward end.

16. The retention system of claim **15**, wherein when the bracket is attached to the body, in the retainer member's second orientation the watercraft accessory is below the hull.

17. The retention system as in claim **15**, wherein the bracket is configured to be attached to the deck and to extend across and outward of a perimeter of the body defined by the forward end, the rearward end, and opposing sides of the body at one of the sides of the body, wherein each opposing side extends between the forward end and the rearward end,

the bracket comprises a collar disposed at least partially outward of the perimeter;

the elongated pole is received by the collar so that the collar at least partially encloses an outer perimeter of the pole,

wherein when the bracket is attached to the body and the collar is attached to the bracket,

in the collar's first orientation, the pole's dimension of elongation is generally parallel to the body's dimension of elongation, and

in the collar's second orientation, the pole's dimension of elongation is generally perpendicular to the body's dimension of elongation.

18. The retention system as in claim **17**, further comprising

a selectively engageable first lock between the collar and the pole, and

a selectively engageable second lock between the bracket and the collar, the second lock being configured to resist movement of the collar out of the first orientation and to resist movement of the collar out of the second orientation.

19. A retention system for use with a watercraft, the watercraft comprising an elongated body comprising a forward end, a rearward end opposite the forward end along the body's dimension of elongation, a hull extending from the forward end to the rearward end and forming a bottom of the body, and a deck extending from the forward end to the rearward end and forming a top of the body, the retention system comprising:

a bracket configured to be attached to the body; and

a retainer member attached to the bracket movably between a first orientation with respect to the bracket and a second orientation with respect to the bracket, the retainer member configured to receive an elongated pole;

wherein when the bracket is attached to the body, the retainer member is attached to the bracket, and the elongated pole is received in the retainer member,

in the retainer member's first orientation, the pole is received by the retainer member so that the pole's dimension of elongation is generally parallel to the body's dimension of elongation, and

in the retainer member's second orientation, the pole's dimension of elongation is generally perpendicular to the body's dimension of elongation.

20. The retention system of claim **19**, further comprising a sonar transducer coupled with a distal end of the elongated pole.

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